Gpyro Verification and Validation Manual

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Contents

Chapter 1

Thermal 1D

1.1 Heat Conduction

Four distinct cases of one-dimensional transient heat conduction through a L =0.1 m thick plate are analyzed. In each case, the plate is initially at a uniform temperature of T_{ini} =20°C and is exposed on one side to ambient air at T_∞ =120°C with convection, while the other side is perfectly insulated. The key parameters for each case are summarized in the table 1.1. The evolution of the temperature profiles can be seen in fig. 1.1.

The Analytical temperature field in the plate is given by Incropera et al [1]:

$$T(z,t) = T_\infty + (T_{ini} - T_\infty) \sum_{n=1}^{\infty} C_n \cos\left(\beta_n \frac{z}{L}\right) \exp\left(-\beta_n^2 \frac{\alpha t}{L^2}\right) \tag{1.1}$$

where $\alpha = k/(\rho c)$ is the thermal diffusivity, $C_n = \dfrac{4\sin(\beta_n)}{2\beta_n + \sin(2\beta_n)}$ a constant, and the eigenvalues β_n satisfy $\beta_n \tan(\beta_n) = Bi$, where Bi is the Biot number.

Table 1.1: Properties for Each Heat Conduction Case

Parameter	**Case A**	**Case B**	**Case C**	**Case D**
Thermal Conductivity, k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	0.1	0.1	1.0	10.0
Density, ρ ($\mathrm{kg\,m^{-3}}$)	100	100	1000	10000
Specific Heat, c ($\mathrm{kJ\,kg^{-1}\,K^{-1}}$)	1000	1000	1000	1000
Convective Coefficient, h $\mathrm{W\,m^{-2}\,K^{-1}}$	100	10	10	10
Biot Number, $Bi = \frac{hL}{k}$	**100**	**10**	**1**	**0.1**

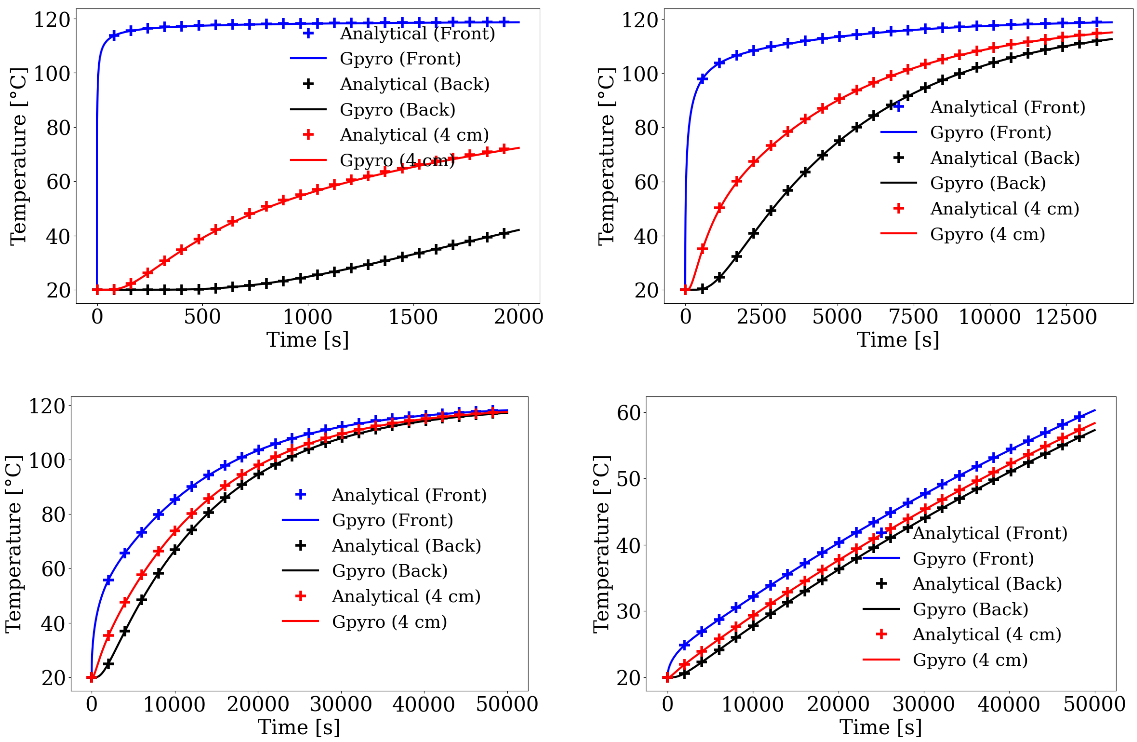


Figure 1.1: Temperature evolution for Case A (top-left), B (top-right), C (bottom-left) and Case D (bottom-right).

1.2 Heat Conduction With Temperature-Dependent Thermal Properties

A one-dimensional heat conduction problem through a planar slab of thickness $L = 1$ cm is considered. The material is initially at a uniform temperature of $T_{\text{ini}} = 0$ °C. The front surface is exposed to a hot gas at 700 °C, while the back surface is exposed to a gas at 0 °C. Heat transfer at both boundaries is governed exclusively by convection, with a constant convective heat transfer coefficient of $h_c = 10\ \text{W} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ (see Fig.1.2). The thermal conductivity k and the specific heat capacity c_p are temperature-dependent. Their temperature evolution is assumed to be linear, and their values at some specific temperatures are reported in Table 1.2. Note that Gpyro models temperature-dependent properties in the form $\phi(T) = \phi_0 \left(\frac{T}{T_{\text{ref}}}\right)^n$, therefore, distinct coefficients ϕ_0 and exponents n are selected such that the resulting variation is locally quasi-linear and consistent with the reference data.

The results are presented in Fig. 1.3, where the numerical results are compared with those obtained using the finite-difference solver HEATING [2, 3].

Table 1.2: Thermophysical properties of the slab material.

Temperature (°C)	k (W m^{-1} K^{-1})	c_p (J kg^{-1} K^{-1})	ρ (kg m^{-3})
0	0.10	1000	10 000
100	0.15	1200	10 000
200	0.20	1000	10 000

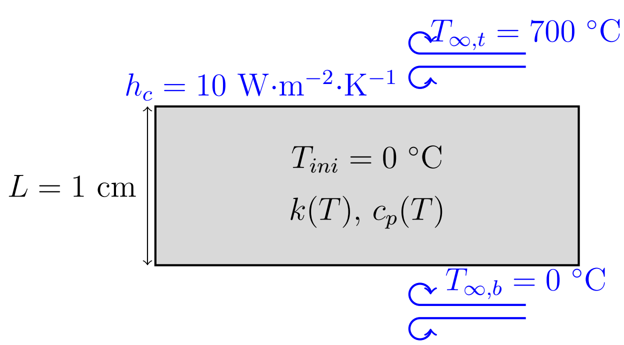


Figure 1.2: Configuration schematic

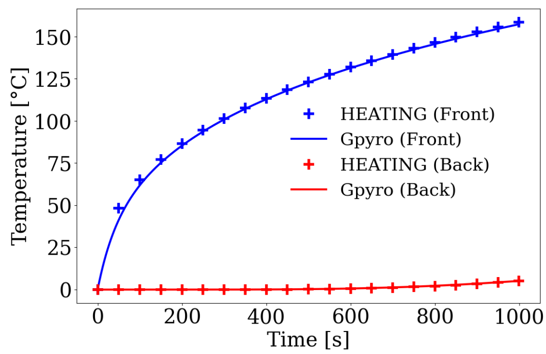


Figure 1.3: Evolution of front and back temperatures with time

1.3 Convective Cooling

This case investigates one-dimensional transient heat conduction in a thick slab of thickness $L = 1$ m. The slab is initially at a uniform temperature of $T_{ini} = 1000$ °C and is exposed to ambient air at $T_\infty = 0$ °C on one face. Heat exchange at the exposed surface occurs by convection and is characterized by a heat transfer coefficient $h_c = 1\ \mathrm{W\,m^{-2}\,K^{-1}}$. The opposite face is assumed to be perfectly insulated. The configuration is illustrated in Figure 1.4. The thermophysical properties of the slab material are summarized in Table 1.3.

The numerical results are presented in Figure 1.5 and are compared with the analytical solution reported by Incropera et al. [1], which follows the same formulation as Equation 1.1.

Table 1.3: Thermophysical properties of the slab material.

k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)
1	1	1000

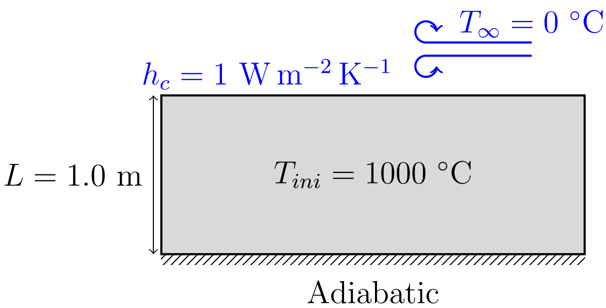


Figure 1.4: Schematic representation of the heat conduction configuration.

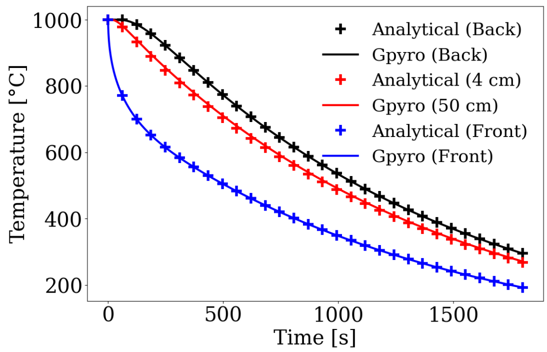


Figure 1.5: Temporal evolution of temperature at different locations within the solid.

1.4 Insulated Multi-Layer Steel Plate

This case considers steady-state one-dimensional heat conduction through a multilayer composite slab composed of a central steel layer of thickness $L_{\text{steel}} = 1$ cm, symmetrically covered by two insulation layers of thickness $L_{\text{ins}} = 2$ cm each. The assembly is subjected to asymmetric convective boundary conditions: the hot-side surface is exposed to ambient air at $T_{\infty,t} = 480$ °C, while the cold-side surface is exposed to air at $T_{\infty,b} = 20$ °C. The slab is heated for a sufficiently long duration (10 hours) to ensure steady-state conditions. The convective heat transfer coefficient is assumed identical on both sides and equal to $h_c = 10\ \mathrm{W\,m^{-2}\,K^{-1}}$ (see Fig. 1.6). The thermophysical properties of the materials are summarized in Table 1.4.

Table 1.4: Thermophysical properties of the materials.

Material	k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)
Steel	50.0	500	7500
Insulation	0.2	1000	200

At steady state, the heat flux is constant through the slab and is given by [1]:

$$q'' = \frac{T_{\infty,t} - T_{\infty,b}}{\frac{1}{h_c} + \frac{L_{\text{ins}}}{k_{\text{ins}}} + \frac{L_{\text{steel}}}{k_{\text{steel}}} + \frac{L_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_c}} \tag{1.2}$$

The temperature field within each layer is piecewise linear between successive interfaces, and the temperatures at the interfaces are expressed as:

$$\begin{cases} T_0 = T_{\infty,t} - \dfrac{q''}{h_c} & : \text{ hot-side surface temperature} \\ T_1 = T_0 - q''\dfrac{L_{\text{ins}}}{k_{\text{ins}}} & : \text{ interface between hot-side insulation and steel} \\ T_2 = T_1 - q''\dfrac{L_{\text{steel}}}{k_{\text{steel}}} & : \text{ interface between steel and cold-side insulation} \\ T_3 = T_2 - q''\dfrac{L_{\text{ins}}}{k_{\text{ins}}} & : \text{ cold-side surface temperature} \end{cases} \tag{1.3}$$

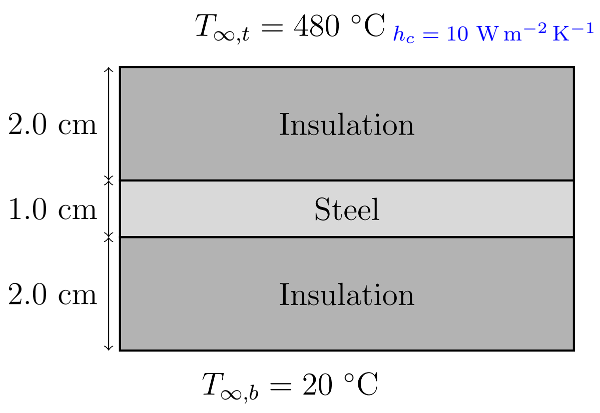


Figure 1.6: Schematic of the insulated steel plate configuration.

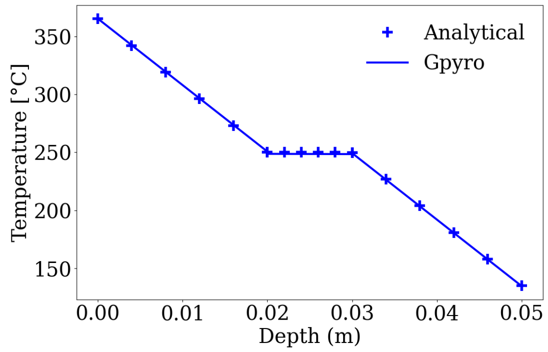


Figure 1.7: Temperature profile as a function of depth within the multilayer slab.

1.5 Fixed temperatures at boundaries

A slab of thickness $L = 10$ cm is considered. Constant temperatures are imposed at the boundaries, with a temperature of $T_{\infty,t} = 400$ °C at the top surface and $T_{\infty,b} = 20$ °C at the bottom surface. The initial temperature of the slab is $T_{ini} = 0$ °C, as illustrated in Figure 1.8. The thermophysical properties of the material used in the simulation are reported in Table 1.5.

The numerical results are validated against the analytical solution given by Incropera et al. [1]. The analytical temperature distribution is expressed as:

$$T(z,t) = T_{\mathrm{ss}}(z) + \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi z}{L}\right) \exp\left[-\alpha\left(\frac{n\pi}{L}\right)^2 t\right] \tag{1.4}$$

where the steady-state temperature profile is

$$T_{\mathrm{ss}}(z) = T_{\infty,t} + \frac{T_{\infty,b} - T_{\infty,t}}{L} z \tag{1.5}$$

and the Fourier coefficients are given by

$$C_n = \frac{2}{n\pi}\left[(T_{ini} - T_{\infty,t}) + (T_{ini} - T_{\infty,b})(-1)^{n+1}\right] \tag{1.6}$$

Here, T_0 denotes the initial uniform temperature of the slab, and $\alpha = k/(\rho c_p)$ is the thermal diffusivity.

A comparison between numerical and analytical temperature profiles is presented in Figure 1.9.

Table 1.5: Thermophysical properties of the slab material

k (W m^{-1} K^{-1})	c_p (J kg^{-1} K^{-1})	ρ (kg m^{-3})	ε (–)
0.10	1000	100	0.0

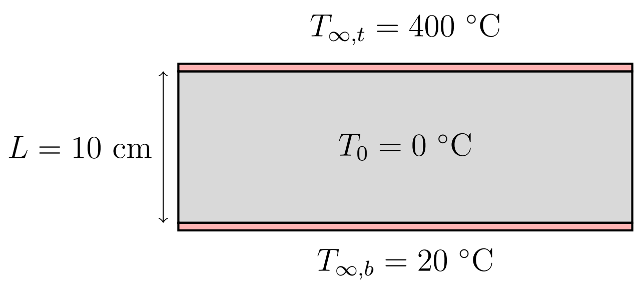


Figure 1.8: Schematic of the slab with fixed temperature boundary conditions

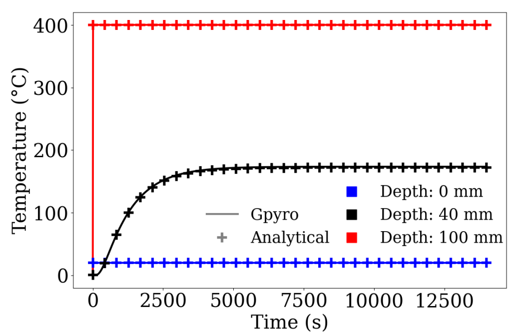


Figure 1.9: Comparison between numerical and analytical temperature evolution

1.6 Radiative heat losses

This verification case investigates a hot plate losing heat solely through thermal re-radiation, which is modeled using the Stefan–Boltzmann law:

$$q''_{\mathrm{r,out}} = \varepsilon\sigma\left(T_s^4 - T_{\infty,t}^4\right) \tag{1.7}$$

where T_s denotes the surface temperature, ε is the surface emissivity, and σ is the Stefan–Boltzmann constant. The key parameters associated with each subcases are summarized in Table 1.6.

Two subcases are considered, corresponding to different ambient temperatures: $T_{\infty,t} = 0$ °K and $T_{\infty,t} = 273.15$ °K. In both configurations, the plate is initially at a uniform temperature of $T_0 = 500$ °K. One surface of the plate is assumed to be adiabatic, while the back surface is exposed to the surroundings. A schematic representation of the computational configuration is provided in Figure 1.10.

The temporal evolution of the temperature profiles is presented in Figure 1.11. Since no analytical solution is available for this configuration, the numerical results are compared against simulations performed with FDS.

Table 1.6: Thermophysical properties

Case	T_∞ (K)	k (W m^{-1} K^{-1})	c_p (kJ kg^{-1} K^{-1})	ρ (kg m^{-3})	$\epsilon(-)$
1	0	0.186	1.764	380	1.0
2	273.15	0.186	1.764	380	1.0

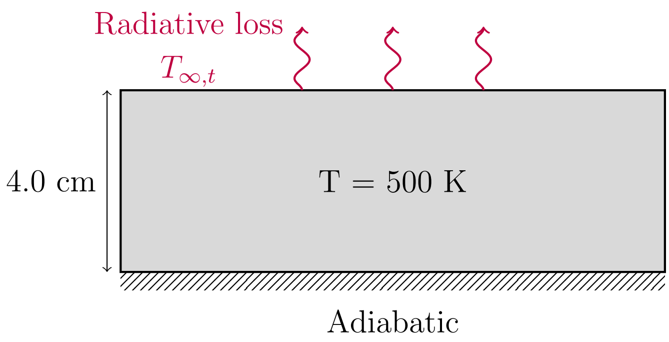


Figure 1.10: Configuration schematic

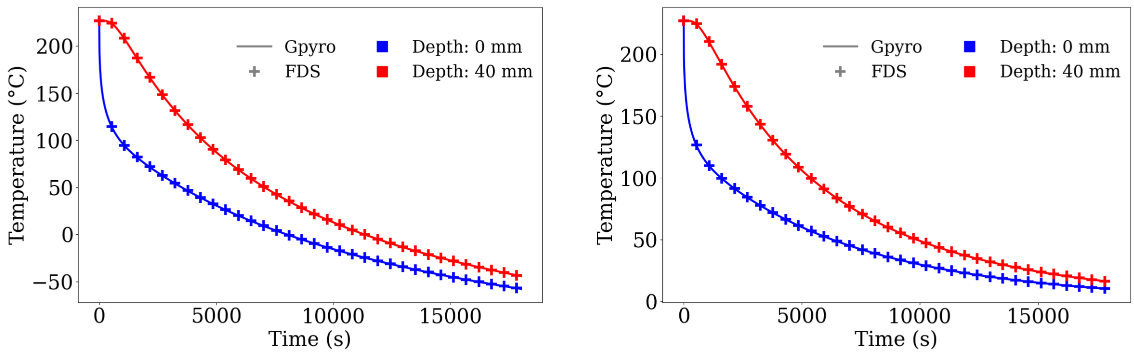


Figure 1.11: Temperature evolution for Case 1 (left) and 2 (right).

1.7 Fixed heat flux boundary condition

This verification case examines one-dimensional heat conduction in a slab subjected to a prescribed surface heat flux. A slab of thickness 4 cm is initially at a uniform temperature of $T_{ini} = 27$ °C. The exposed surface is subjected to a constant incident heat flux of $q''_{\text{ext}} = 2\ \text{kW}\,\text{m}^{-2}$. The material is assumed to be opaque, such that no in-depth radiation penetration occurs. The opposite surface is considered adiabatic, as illustrated in Figure 1.12. The thermophysical properties of the material are summarized in Table 1.7.

The numerical results are validated against the analytical solution for transient heat conduction in a semi-infinite slab subjected to a constant surface heat flux, as presented by Incropera et al. [1]. The analytical solution is given by:

$$T(z,t) - T_{ini} = \frac{2\varepsilon q''_{\text{ext}}}{k}\left(\frac{\alpha t}{\pi}\right)^{1/2} \exp\left(-\frac{z^2}{4\alpha t}\right) - \frac{\varepsilon q''_{\text{ext}} z}{k}\,\text{erfc}\left(\frac{z}{2\sqrt{\alpha t}}\right) \tag{1.8}$$

where $\alpha = k/(\rho c_p)$ is the thermal diffusivity, ε is the surface emissivity, and x denotes the distance from the exposed surface.

Table 1.7: Thermophysical properties used in the simulation

k ($\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$)	c_p ($\text{kJ}\,\text{kg}^{-1}\,\text{K}^{-1}$)	ρ ($\text{kg}\,\text{m}^{-3}$)	ε (−)
0.186	1.764	380	1.0

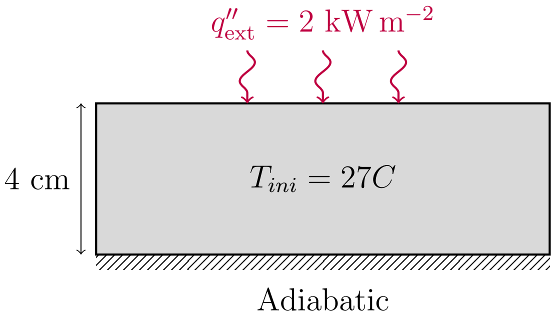


Figure 1.12: Schematic representation of the computational configuration.

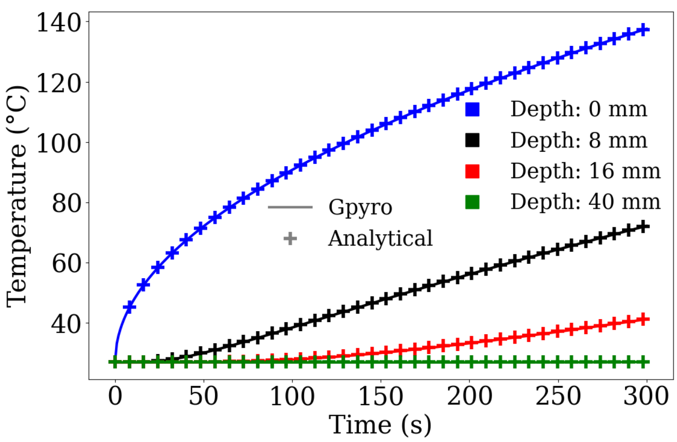


Figure 1.13: Temperature evolution at different depths inside the slab.

1.8 In-depth radiation absorption

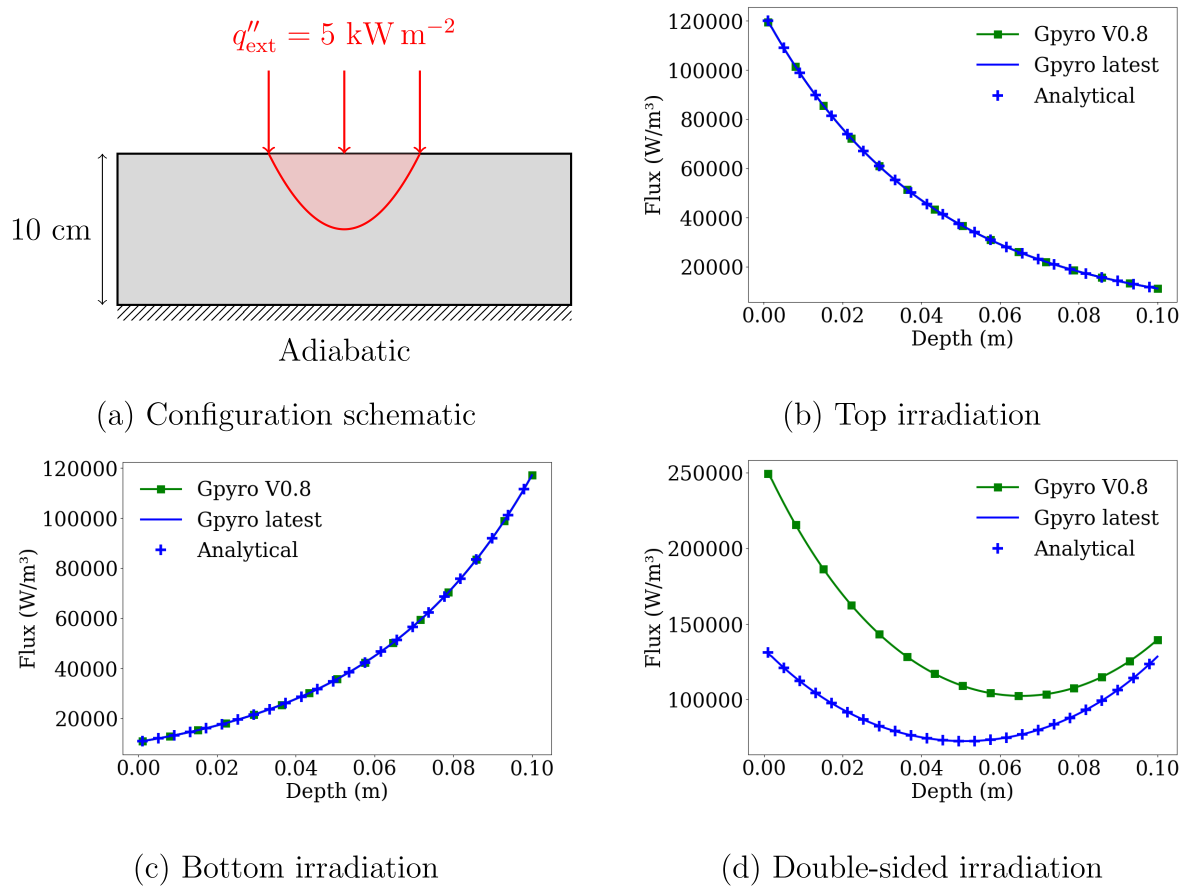
This verification case is designed to assess the capability of Gpyro to accurately model one-dimensional radiative heat absorption in a semi-opaque material. A slab of thickness 10 cm is subjected exclusively to an incident radiative heat flux of $q''_{\mathrm{ext}} = 5\ \mathrm{kW\,m^{-2}}$, which penetrates the material. The slab is assumed to be semi-opaque and characterized by a constant absorption coefficient $\kappa = 24\ \mathrm{m^{-1}}$. Under these assumptions, the volumetric radiative heat source resulting from in-depth radiation absorption is described by an Beer-Lambert law:

$$\dot{Q}''_r(z) = \varepsilon \kappa q''_{\mathrm{ext}} \exp\left(-\kappa z\right) \tag{1.9}$$

where z denotes the distance from the irradiated surface and ε is the surface emissivity. Three configurations are considered: radiation applied at the top surface, at the bottom surface, and at both surfaces simultaneously. For each case, numerical results (annotated Gpyro latest) are compared with the corresponding analytical solution and with results obtained using the original version of `Gpyro` (annotated Gpyro V0.8). The thermophysical properties and simulation parameters are listed in Table 1.8.

Table 1.8: Simulation properties

k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)	ε ($-$)	κ ($\mathrm{m^{-1}}$)	q''_{ext} ($\mathrm{kW\,m^{-2}}$)
0.10	1000	100	1.0	24	5



(a) Configuration schematic

(b) Top irradiation

(c) Bottom irradiation

(d) Double-sided irradiation

Figure 1.14: Comparison between Gpyro and analytical solutions for in-depth radiation.

1.9 In-Depth Radiation With Surface Convection

This verification case assesses the ability of Gpyro to represent one-dimensional heat absorption due to volumetric radiation combined with surface cooling by convection. A slab of thickness $L = 10$ cm is considered.The thermophysical properties of the material are summarized in Table 1.9. The top surface is subjected to convective heat transfer with a constant convective heat transfer coefficient of $h_c = 10\ \mathrm{W\,m^{-2}\,K^{-1}}$ and an ambient gas temperature of $T_\infty = 25$ °C. In addition, the top surface is also exposed to an external radiative heat flux of $q''_{\mathrm{ext}} = 5\ \mathrm{kW\,m^{-2}}$.The material is assumed to be semi-transparent, characterized by an absorption coefficient of $\kappa = 24\ \mathrm{m^{-1}}$. The bottom surface of the slab is assumed to be adiabatic. The initial temperature is uniform throughout the material, with $T_{ini} = 25$ °C. The configuration is illustrated in Fig. 1.16.

An analytical solution for transient temperature evolution within the slab is available and is taken from the formulation presented in Lautenberger's thesis [4]:

$$
\begin{aligned}
T(z) - T_{ini} = {} & \frac{2\varepsilon q''_{\mathrm{ext}}\kappa^2}{k} \sum_{m=1}^{\infty} \frac{1}{\lambda_m(\kappa^2 + \lambda_m^2)} \\
& \times \frac{\cos(\lambda_m L) + \frac{\lambda_m}{\kappa}\sin(\lambda_m L) - \exp(-\kappa L)}{\lambda_m L + \frac{1}{2}\sin(2\lambda_m L)} \\
& \times \cos\big(\lambda_m(L-z)\big)\left(1 - \exp(-\alpha\lambda_m^2 t)\right)
\end{aligned}
\tag{1.10}
$$

The eigenvalues λ_m are obtained by solving the following equation:

$$
\cot(\lambda_m L) = \frac{k}{h_c}\lambda_m \tag{1.11}
$$

Table 1.9: Material properties

k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)	ε	κ ($\mathrm{m^{-1}}$)
0.10	1000	100	1	24

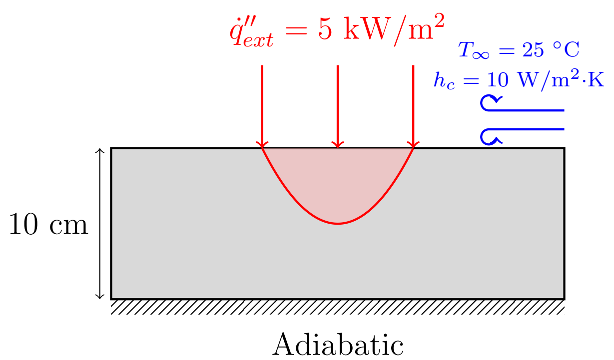


Figure 1.15: Configuration schematic

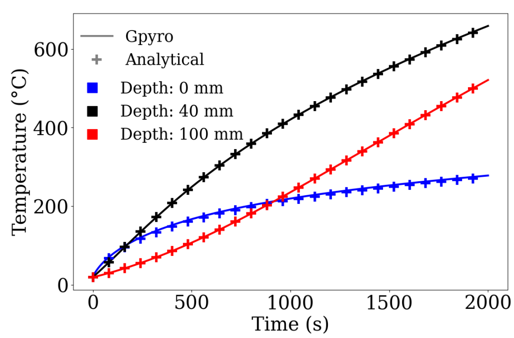


Figure 1.16: Temperature evolution at different points in the slab

1.10 Mixed heat transfer

This verification case assesses the capability of Gpyro to accurately represent 1D heat transfer involving multiple modes of transfer, and compares the results with those from FDS. The scenario consists of a medium-density fibreboard (MDF) sample subjected to a cone calorimeter test in a nitrogen atmosphere. The properties of the simulated materials can be found in table 1.10. The material is exposed to an external heat flux of q''_{ext} = 1 kW/m^2 and ambient air at 25 °C, with a convection coefficient of $h_c = 10$ W m^{-2} K^{-1}.

Table 1.10: Thermophysical properties of MDF

k (W m^{-1} K^{-1})	c_p (J kg^{-1} K^{-1})	ρ (kg m^{-3})	$\varepsilon(-)$
0.15	1340	605	0.86

The results obtained using Gpyro are compared to those from FDS. Figure 1.18 presents the comparison of the temperature evolution in time at various depths.

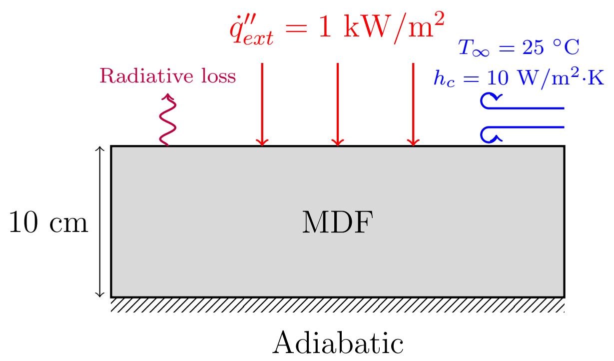


Figure 1.17: Configuration schematic

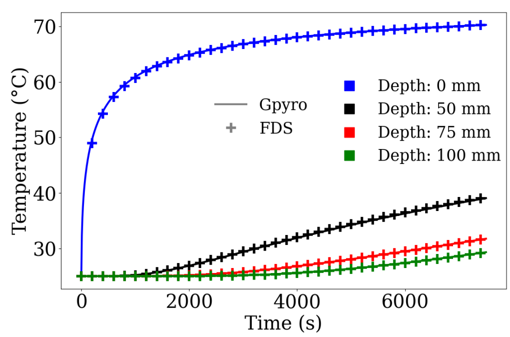


Figure 1.18: Temperature evolution comparison

Chapter 2

Thermal 2D-3D

2.1 Thermal equilibrium in a heterogeneous 3D block

This verification case evaluates the ability of the thermal solver to conserve energy in a closed system. A cubic domain of $V_{tot} = 20\times20\times0.20$ cm is divided into two regions: a hot sub-cube of 10 cm side located at one corner, with an initial temperature $T_{\text{hot}} = 600°\text{C}$, and the remaining volume at $T_{\text{cold}} = 20°\text{C}$ as shown in Figure 5.6. The two regions have distinct thermophysical properties, given in Table 5.2. All boundaries are adiabatic, ensuring no external heat exchange. The material is anisotropic to increase complexity. The theoretical equilibrium temperature T_{eq} is obtained from total energy conservation:

$$H_{\text{hot}} = \rho_{\text{hot}} c_{p,\text{hot}} V_{\text{hot}} T_{\text{hot}} \ ; \ H_{\text{cold}} = \rho_{\text{cold}} c_{p,\text{cold}} V_{\text{cold}} T_{\text{cold}} \ ; \ H_{\text{total}} = H_{\text{hot}} + H_{\text{cold}} \tag{2.1}$$

$$T_{\text{eq}} = \frac{H_{\text{total}}}{\rho_{\text{hot}} c_{p,\text{hot}} V_{\text{hot}} + \rho_{\text{cold}} c_{p,\text{cold}} V_{\text{cold}}} \approx 116.45°\text{C} \tag{2.2}$$

The time–temperature evolution at three points of the domain is shown in Figure 2.2, and Figure 2.3 shows a slice view at $x = 5$ cm at different time steps. The effect of anisotropic conductivity is visible, as the heat propagation rate is higher in one direction.

Table 2.1: Thermophysical properties of the two regions.

Region	T_{ini} (°C)	k_x	k_y	k_z $(\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1})$	c_p $(\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1})$	ρ $(\text{kg}\cdot\text{m}^{-3})$
Hot	400	8	5	10	1000	1000
Cold	20	8	5	10	700	600

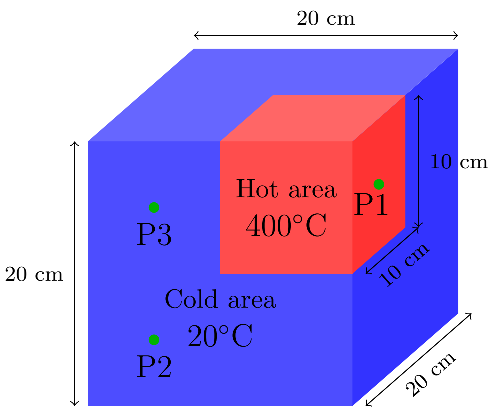


Figure 2.1: Domain configuration

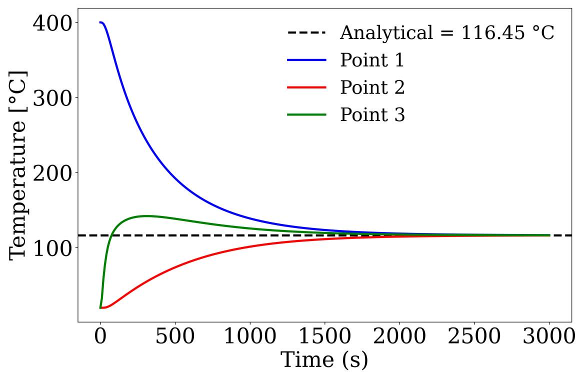


Figure 2.2: Temperature evolution.

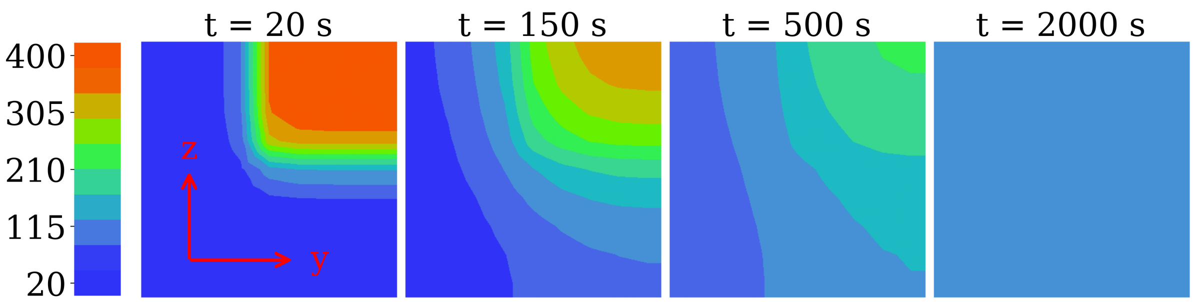


Figure 2.3: Temperature distribution on the $x = 5$ cm slice at different time steps.

2.2 3-D Heat Diffusion in a Steel I-Beam

This case is intended to verify the Gpyro code under three-dimensional heat diffusion conditions involving a complex I-beam geometry. A steel I-beam is considered, with an overall square cross-section of 0.4 m on each side. The flanges are 6 cm thick, while the web has a thickness of 4 cm, as shown in Figure 2.4.

The spatial grid resolution used in both the Gpyro and FDS simulations is $\Delta x = \Delta y = \Delta z = 1$ cm. The thermal properties of the steel are assumed to be constant, with a thermal conductivity $k = 45\ \mathrm{W\,m^{-1}\,K^{-1}}$, a density $\rho = 7850\ \mathrm{kg\,m^{-3}}$, and a specific heat capacity $c = 0.60\ \mathrm{kJ\,kg^{-1}\,K^{-1}}$.

All boundary conditions are adiabatic, except for a heated patch located on the front half of the bottom flange, which is maintained at a constant temperature of 800 °C. The initial temperature of the steel is set to 20 °C. The simulation is run for a total duration of 3600 s.

Temperatures are monitored at six locations within the beam, as indicated in Figure 2.4. The temporal evolution of the temperature at these points is presented in Figure 2.5. In addition, temperature profiles are extracted along the x and z directions at different times and are shown in Figure 2.6.

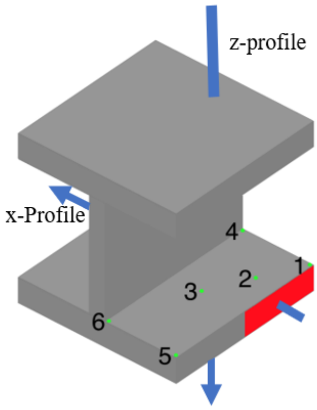


Figure 2.4: I-beam geometry [2]

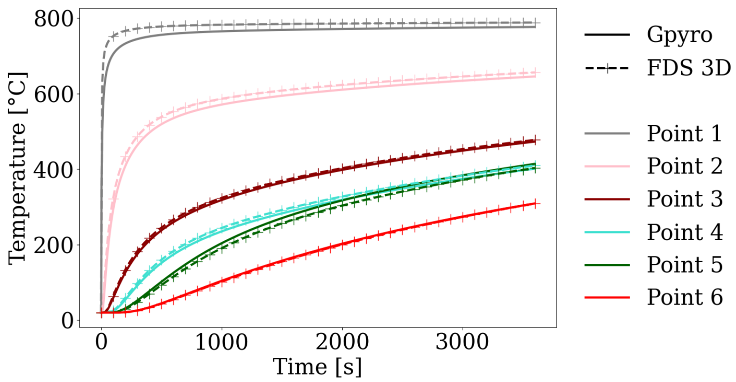


Figure 2.5: Time evolution of Temperature at the measurement points.

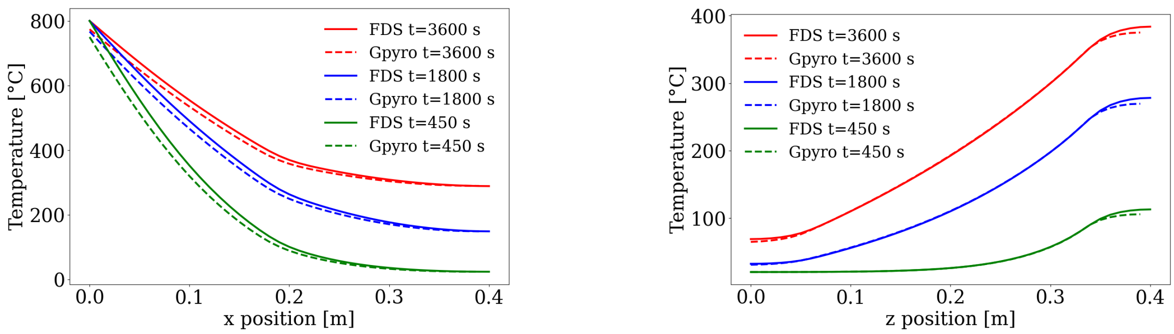


Figure 2.6: Temperature profiles along the x direction (left) and along the z direction (right) at different times.

Chapter 3

Species and Mass Conservation

3.1 Mixture law

This verification case evaluates the ability of GPYRO to apply the mixture law for reacting condensed-phase materials. For a generic property ϕ (such as bulk density ρ, porosity Ψ, thermal conductivity k or the emissivity ε), the effective mixture value is written as:

$$\bar{\phi}(x,t) = \sum_{i=1}^{M} X_i(x,t)\,\phi_i, \tag{3.1}$$

where X_i is the volume fraction of material i.

A simulation is performed in which material A is converted into material B at a constant reaction rate $r = 0.001\ \mathrm{kg\,m^{-3}\,s^{-1}}$. The material properties are summarized in Table 3.1. The analytical solution for the mass and volume fractions Y is given by:

$$\begin{cases} Y_A(t) = \max(1 - rt,\ 0), \\ Y_B(t) = 1 - Y_A(t), \end{cases} \qquad \begin{cases} X_A(t) = \bar{\rho}\, Y_A/\rho_A, \\ X_B(t) = \bar{\rho}\, Y_B/\rho_B, \end{cases} \qquad \text{with } \bar{\rho}(t) = \left(\frac{Y_A}{\rho_A} + \frac{Y_B}{\rho_B}\right)^{-1} \tag{3.2}$$

Table 3.1: Thermophysical properties used for the mixture law verification

Material	k ($\mathrm{W\,m^{-1}\,K^{-1}}$)	ρ ($\mathrm{kg\,m^{-3}}$)	ρ_s ($\mathrm{kg\,m^{-3}}$)	c_p ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	ε (–)
Material A	1.0	1000	2000	1200	1.0
Material B	0.5	600	800	1900	0.0

The temporal evolution of mass fractions, volume fractions, and effective properties is shown in Figure 3.1. Numerical results obtained with the current version of GPYRO are compared to the analytical solution and to results from GPYRO V0.8, highlighting the correct implementation of the mixture law in the version of GPYRO.

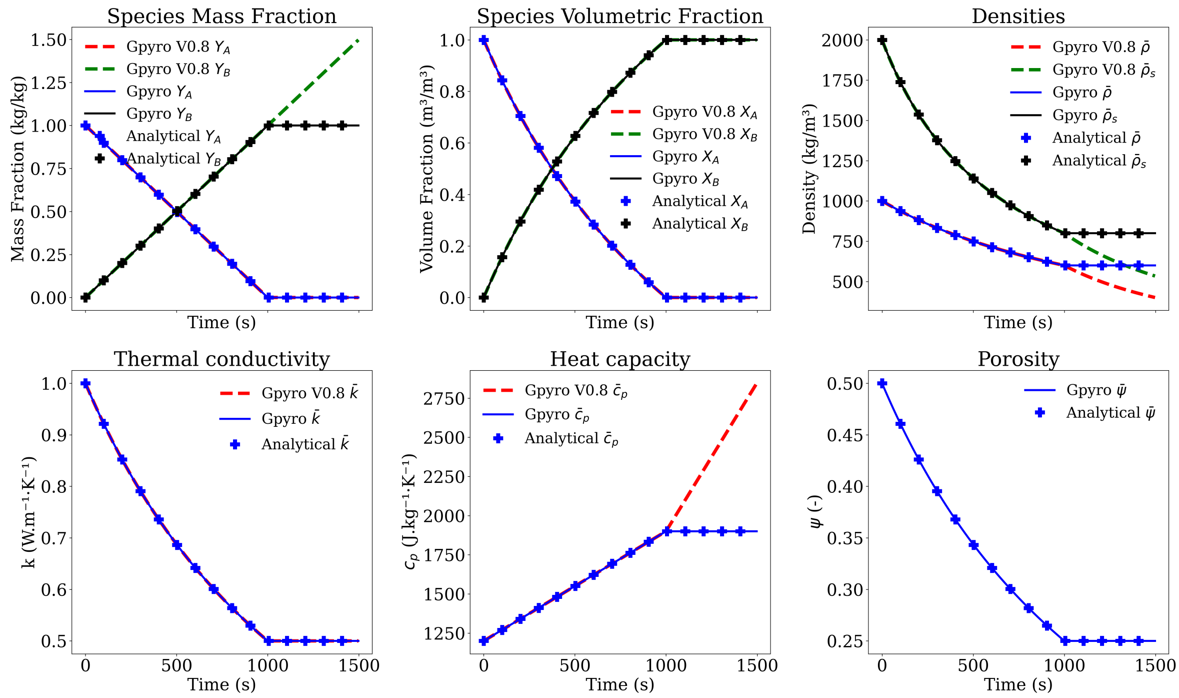


Figure 3.1: Mixture law verification: comparison between analytical and numerical results

3.2 Thermophysical properties temperature dependancy

This verification case assesses the ability of GPYRO to correctly evaluate temperature-dependent thermophysical properties. A one-dimensional simulation is performed in which the temperature increases linearly with time at a constant heating rate of 1 $\mathrm{K\,s^{-1}}$. The material properties are prescribed as explicit functions of temperature according to the following relations:

$$c_p(T) = c_{p,0}\left(\frac{T}{T_{\mathrm{ref}}}\right)^{n_c} \tag{3.3}$$

$$k(T) = k_0\left(\frac{T}{T_{\mathrm{ref}}}\right)^{n_k} + \sigma\,\gamma\,T^3 \tag{3.4}$$

$$\rho(T) = \rho_0\left(\frac{T}{T_{\mathrm{ref}}}\right)^{n_r} \tag{3.5}$$

where $T_{\mathrm{ref}} = 300$ K is the reference temperature, σ is the Stefan–Boltzmann constant, and the material parameters are listed in Table 3.2.

Table 3.2: Material parameters used for the temperature-dependent properties verification.

k_0 ($\mathrm{W\,m^{-1}\,K^{-1}}$)	n_k ($-$)	ρ_0 ($\mathrm{kg\,m^{-3}}$)	n_r ($-$)	$c_{p,0}$ ($\mathrm{J\,kg^{-1}\,K^{-1}}$)	n_c ($-$)	γ ($\mathrm{m^{-1}}$)
0.2	0.6	1430	0.25	1550	0.3	0.002

The thermophysical properties predicted by GPYRO are compared against the analytical expressions above as the temperature increases. Results obtained with the original implementation (GPYRO V0.8) are also included, as discrepancies are identified for the original version of GPYRO.

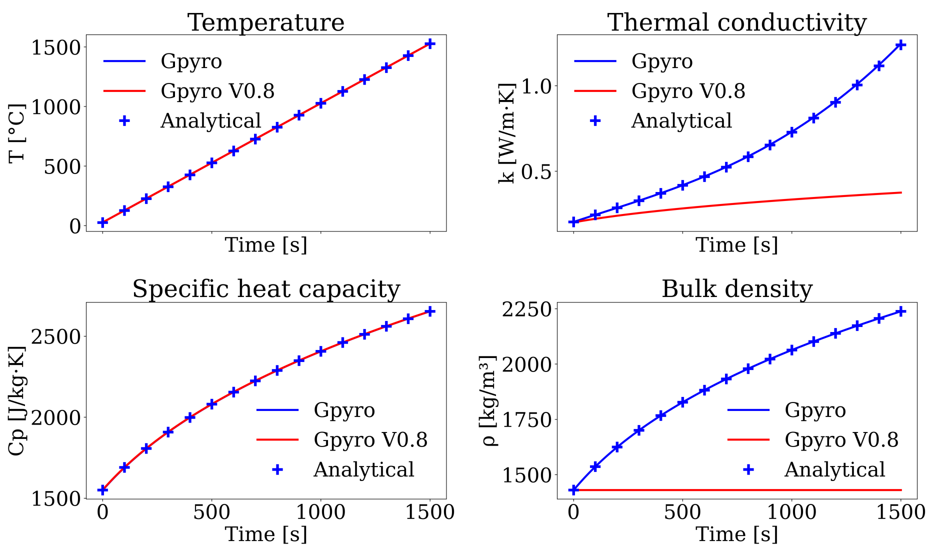


Figure 3.2: Comparison of numerical and analytical temperature-dependent properties.

3.3 Simple TGA

This verification case assesses the ability of Gpyro to accurately reproduce thermogravimetric analysis (TGA) results using a single-step, first-order reaction model. A simple non-charring material undergoing a single degradation reaction is considered. The reaction properties are summarized in Table 3.3.

The simulation is performed in a zero-dimensional configuration, equivalent to a TGA experiment. The initial sample mass is set to $m_{A,0} = 5$ mg, and the temperature is increased at a constant heating rate of $\beta = 5\,\mathrm{K\,min^{-1}}$, starting from an initial temperature of $T_0 = 27°\mathrm{C}$. The total simulation time is 5000 s.

The resulting mass loss rate (MLR) obtained from the Gpyro simulation is compared with the corresponding analytical solution. The analytical formulation of this problem is derived in Appendix B and is recalled here for completeness.

$$\mathrm{MLR}(t) = m_{A,0}\, Z \exp\left(-\frac{E}{R\,(T_0 + \beta t)}\right) \exp\left(-\int_{t_0}^{t} Z \exp\left(-\frac{E}{R\,(T_0 + \beta t')}\right) dt'\right) \tag{3.6}$$

Table 3.3: Reaction properties.

Reaction	Z ($\mathrm{s^{-1}}$)	E ($\mathrm{kJ\,mol^{-1}}$)	n (-)
Material A $\rightarrow$ gases	0.87×10^{13}	163	1

The results obtained using Gpyro are compared to the analytical predictions. Figure 3.3 presents the mass loss rate as a function of time for both approaches.

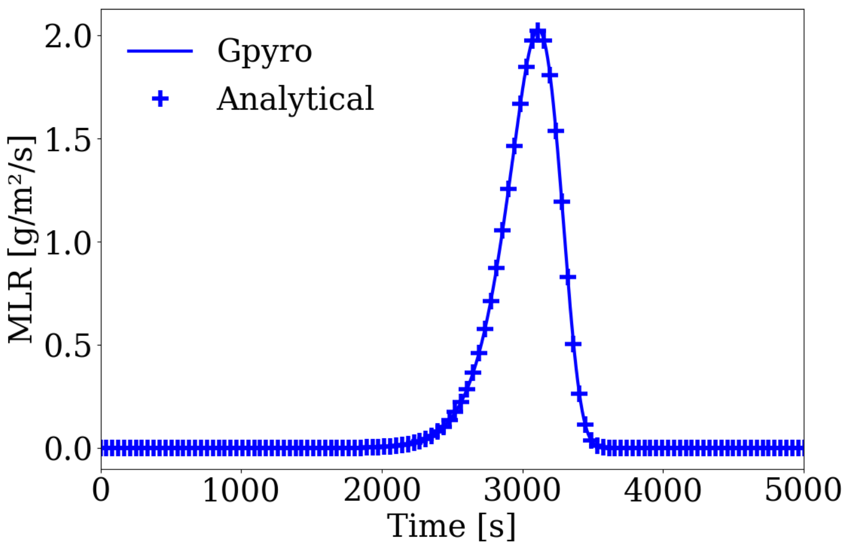


Figure 3.3: Comparison between Gpyro and analytical mass loss rate predictions.

3.4 Oxidation

This verification case assesses the ability of GPYRO to accurately simulate oxidative reactions. A zero-dimensional simulation is performed on a $m_{A,0} = 5$ mg sample composed of a single condensed-phase material A undergoing a one-step oxidation reaction. The material is initially at $T_0 = 20°\mathrm{C}$ and is heated with a constant temperature ramp of $\beta = 5\ \mathrm{K\,min^{-1}}$. The reaction is written as:

$$\mathrm{A} + \theta_{O_2}\,\mathrm{O_2} \longrightarrow \theta\,\mathrm{Char} + (1-\theta+\theta_{O_2})\,\mathrm{G}, \tag{3.7}$$

where $\theta = 0.8$ denotes the char yield and $\theta_{O_2} = 0.2$ the oxygen stoichiometric coefficient.

The reaction is assumed to be first order and is characterized by a pre-exponential factor $Z = 7 \times 10^6\ \mathrm{s^{-1}}$ and an activation energy $E = 190\ \mathrm{kJ\,mol^{-1}}$. The influence of oxygen concentration is investigated by performing simulations with different initial oxygen mass fractions, $Y_{O_2} = 0$, 4.3, 8.2, and 20.5 %.

An analytical solution for this configuration is derived following the same methodology as presented in Appendix B. The resulting mass loss rate (MLR) is directly proportional to the oxygen mass fraction and is given by

$$\mathrm{MLR}(t) = Y_{O_2}\,(1-\theta)\,m_{A,0}\,Z\exp\left(\frac{-E}{R\,(T_0+\beta t)}\right)\exp\left[-\int_{t_0}^{t} Z\exp\left(\frac{-E}{R\,(T_0+\beta t')}\right)\mathrm{d}t'\right] \tag{3.8}$$

The numerical results obtained with GPYRO are compared with the analytical solution in Figure 3.4.

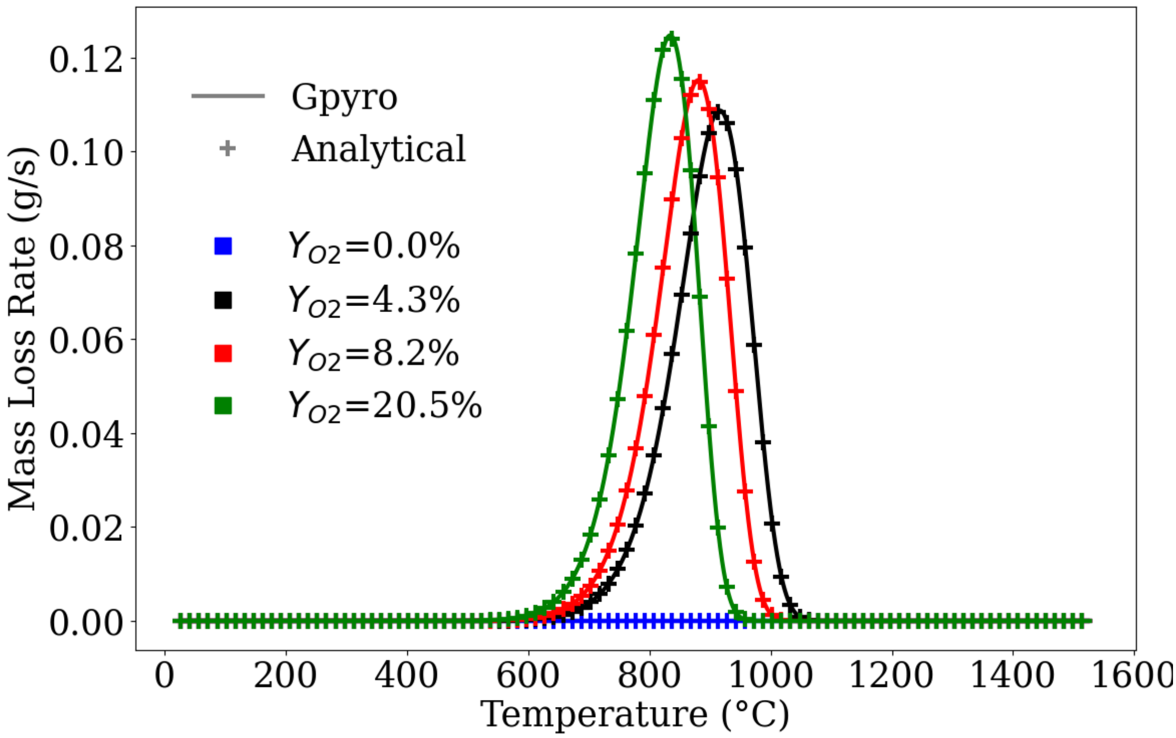


Figure 3.4: Comparison between Gpyro and analytical mass loss rates (MLR).

3.5 Drying and Mass Conservation

This verification case evaluates the ability of GPYRO to ensure mass conservation during a coupled drying and pyrolysis process. A zero-dimensional simulation is performed on an initial 1 kg sample of wet solid containing 20% moisture by mass. The sample is subjected to a constant heating rate of 1 $\mathrm{K\,min^{-1}}$.The process is modeled using two successive reactions. The first reaction represents drying, in which the wet solid releases water:

$$\mathrm{Solid_{wet}} \;\rightarrow\; \theta_1\,\mathrm{Solid_{dry}} + (1-\theta_1)\,\mathrm{H_2O} \tag{3.9}$$

followed by a one-step pyrolysis reaction of the dry solid:

$$\mathrm{Solid_{dry}} \;\rightarrow\; \theta_2\mathrm{Char} + (1-\theta_2)\mathrm{Pyrolysate} \tag{3.10}$$

The kinetic parameters of the reactions are given in Table 3.4.

The expected analytical outcome consists of 0.20 kg of water, 0.48 kg of char, and the remaining mass converted into pyrolysis gases, thereby conserving the initial 1 kg of material.

Table 3.4: Reaction parameters for the drying and pyrolysis processes.

Reaction	Z ($\mathrm{s^{-1}}$)	E ($\mathrm{kJ\,mol^{-1}}$)	T_{crit} (K)	n (–)	θ (–)
Eq. (3.9)	8.04×10^{7}	80	373.15	1	0.8
Eq. (3.10)	1.97×10^{15}	188	–	1	0.6

The temporal evolution of the individual mass contributions predicted by GPYRO is compared against the analytical solution in Figure 3.5, demonstrating correct mass conservation throughout the simulation.

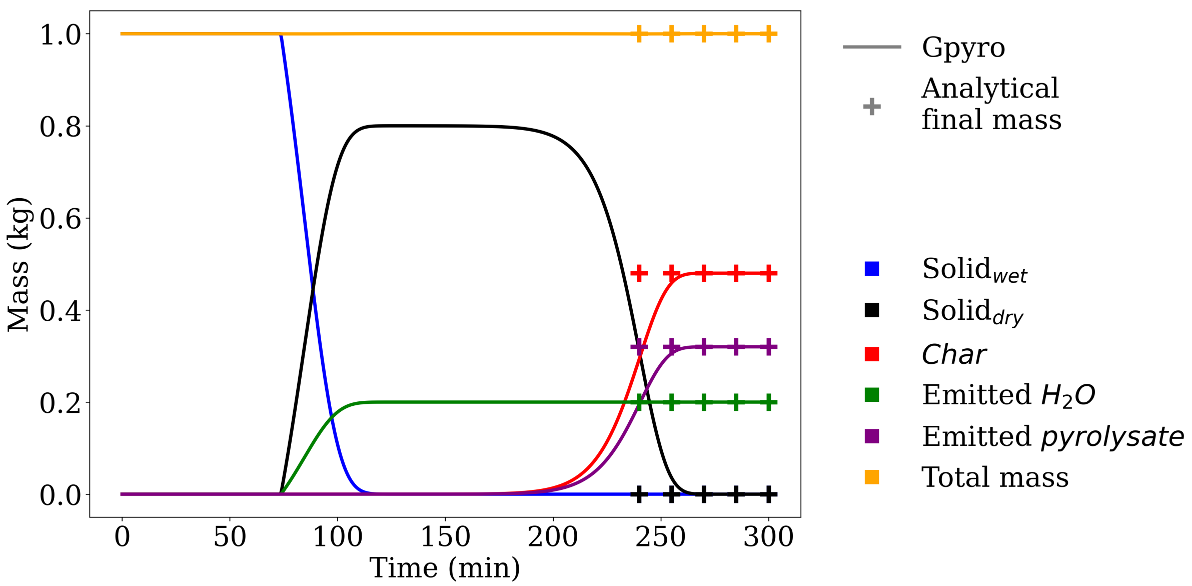


Figure 3.5: Comparison of GPYRO predictions with analytical mass balance results for the drying verification case.

3.6 Reaction Enthalpy

This case is designed to verify the implementation of reaction enthalpy in the code. The volumetric chemical heat source term, denoted $\dot{Q}'''_{sc}$ is expressed as

$$\dot{Q}'''_{sc} = -\sum_{k}^{K} \dot{\omega}'''_{r,k} \, \Delta H_{r,k} + \sum_{i}^{M} \dot{\omega}'''_{net,i} \, h_i \tag{3.11}$$

where K is the number of reactions, $\dot{\omega}'''_{r,k}$ is the reaction rate of reaction k, $\Delta H_{r,k}$ is the reaction enthalpy, $\dot{\omega}'''_{net,i}$ is the net production rate of condensed species i, and h_i is the specific sensible enthalpy of species i.

A one-dimensional simulation is performed, consisting of a single condensed material, denoted material A. This material undergoes a single reaction transforming it into material B without producing gaseous species. The reaction is first order, with an activation energy of $E = 150\,\mathrm{kJ\,mol^{-1}}$ and a pre-exponential factor of $5\times10^{13}\,\mathrm{s^{-1}}$. Two sub-cases are considered. The first corresponds to an endothermic reaction with $\Delta H_k = 1\times10^6\ \mathrm{J\,kg^{-1}}$, while the second corresponds to an exothermic reaction with $\Delta H_k = -1\times10^5\ \mathrm{J\,kg^{-1}}$. A third case is also investigated, in which the reaction is slightly modified to produce gaseous species. In this configuration, 60% of the mass of material A is converted into gases, and the reaction enthalpy is set to $\Delta H_k = 0$. As a result, the first term in Eq. 3.11 vanishes, and only the second term, corresponding to the variation of sensible enthalpy of the condensed phase, contributes to the heat source term.

The numerical Chemical source term and their comparison with analytical solutions are presented in Figure 3.6 for the endothermic case, Figure 3.7 for the exothermic case, and Figure 3.8 for the sensible enthalpy variation case. The analytical solutions are computed directly from Eq. 3.11, using reaction rates extracted from the numerical solution.

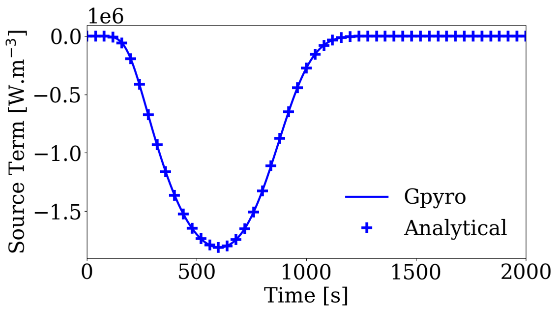


Figure 3.6: Endothermic reaction case.

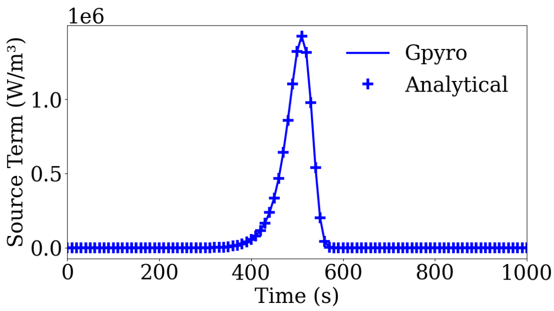


Figure 3.7: Exothermic reaction case.

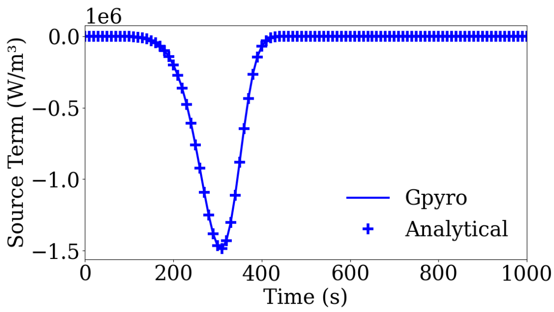


Figure 3.8: Sensible enthalpy case.

3.7 Deformation

A one-dimensional simulation is performed on a slab with an initial thickness $L = 1$ cm. The slab is composed of a single condensed-phase material, A, undergoing a single degradation reaction in which a fraction θ (kg/kg) of the initial mass is converted into a solid product B, while the remaining fraction $(1-\theta)$ is released as gaseous species G, i.e.,$\mathrm{A} \rightarrow \theta\,\mathrm{B} + (1-\theta)\,\mathrm{G}$.

the slab is exposed to a constant external radiative heat flux of 100 $\mathrm{kW\,m^{-2}}$ until it is fully consumed by the reaction.

The deformation factor D is defined as the ratio of the final thickness to the initial thickness. Analytically, at the end of the reaction, it is given by (see Gpyro user guide [5]):

$$D = \frac{\rho_A}{\rho_B}\,\theta, \tag{3.12}$$

where ρ_A and ρ_B are the bulk densities of materials A and B. In Gpyro, the stoichiometric coefficient θ is prescribed indirectly via the user parameter χ as $\theta = 1 + (\rho_B/\rho_A - 1)\chi$.

Five configurations are considered, corresponding to different reaction behaviors obtained by varying the material densities and the parameter χ. The first case corresponds to a purely condensed-phase reaction that does not produce gases, with deformation depending solely on the product and reactant densities. The four other reactions involve gas production: a reaction that does not induce deformation, a shrinking reaction, a swelling reaction, and finally a non-charring material for which the reaction produces only gases. The parameters and expected deformation for each case are summarized in Table 3.5.

Table 3.5: parameters and expected deformation for the five verification cases.

Case	Description	ρ_A $(\mathrm{kg\,m^{-3}})$	ρ_B $(\mathrm{kg\,m^{-3}})$	χ (-)	θ(-)	D (-)
1	Pure condensed-phase reaction	1000	2000	0.0	1.00	0.50
2	No deformation	1000	630	1.0	0.63	1.00
3	Swelling	1000	400	1/3	0.80	2.00
4	Shrinking	1000	600	1.5	0.40	0.67
5	Non-charring	1000	-	1.0	0	0

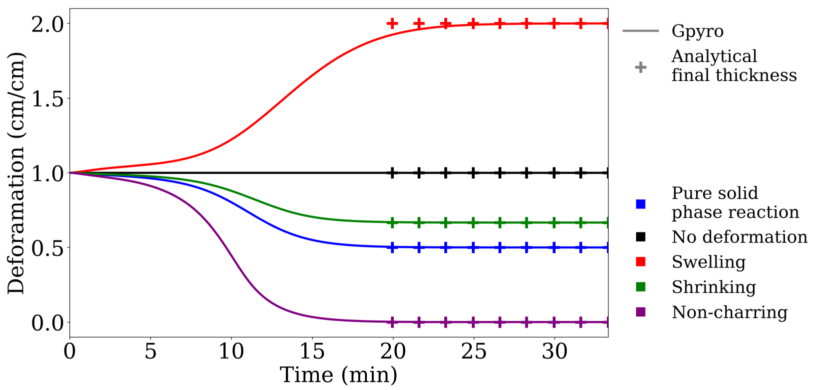


Figure 3.9: Deformation evolution compared to analytical solutions for all cases

Chapter 4

Gas Transport

4.1 Instantaneous gas release rate

This verification case evaluates the behaviour of the instantaneous gas release model. As shown in Figure 4.1, a one-dimensional sample of thickness $L = 10$ cm is considered. A bottom gas influx of $\dot{m}''_{\mathrm{g,ini}} = 0.1\mathrm{kg} \cdot \mathrm{m}^{-2} \cdot \mathrm{s}^{-1}$ is imposed at the lower boundary. The sample undergoes controlled degradation that generates a constant volumetric production of volatile gases $\dot{\omega}'''_{fg} = 0.4\ \mathrm{kg} \cdot \mathrm{m}^{-3} \cdot \mathrm{s}^{-1}$ until complete consumption of the solid (at $t = 1000$ s). These conditions are obtained using the same reaction mechanism as in the verification case "Mixture Law" (Section 3.1).

Figures 4.2 and 4.3 display respectively (i) the total mass flux and total mass loss rate, and (ii) the depth profile of the gas mass flux at $t = 500$ s. The total mass flux corresponds to the sum of the intrinsic mass loss rate and the imposed inflow. Because the vertical axis in Gpyro is directed downward, upward gas motion appears as negative in sign.

Since the gas production rate is constant throughout the domain, the analytical mass-flux distribution varies linearly:

$$\dot{m}''_{\mathrm{gas}}(z) = -\dot{m}''_{\mathrm{g,ini}} - \dot{\omega}'''_{fg}(z - L) \tag{4.1}$$

A time step of 1 s and spatial discretisation of $\Delta z = 0.1$ mm are used.

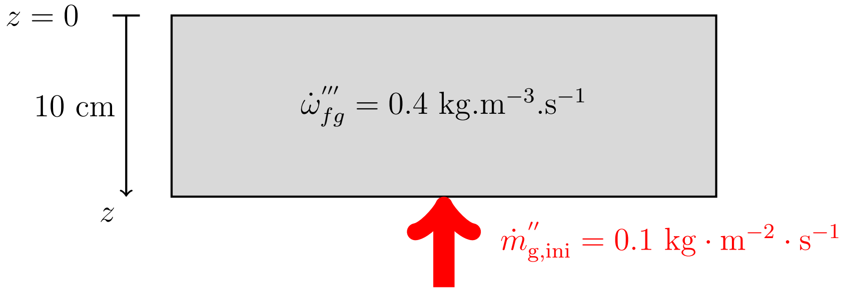


Figure 4.1: Configuration schematic of the instantaneous gas-release test case.

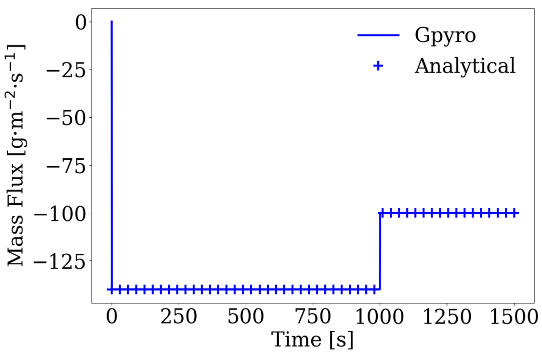


Figure 4.2: Total mass flux and total mass loss rate.

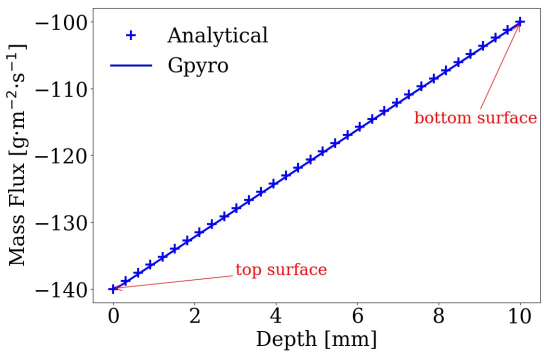


Figure 4.3: Depth profile of mass flux at $t = 500$ s.

4.2 Darcy Flow with Fixed Pressure

This verification case examines the behaviour of the pressure solver in a porous material under steady one-dimensional gas flow with fixed pressure boundary conditions. A uniform porous layer of thickness $L = 10$ cm is considered (Figure 4.4). The temperature is fixed at $T_0 = 300$ K, and the permeability of the material is $\bar{K} = 5 \cdot 10^{-10}$ m^2. The pressure is prescribed at both ends of the domain:

$$P(0) = P_0 = 1 \text{ bar}, \qquad P(L) = P_L = 2 \text{ bar}.$$

The gas kinematic viscosity is computed from the Chapman–Enskog model. In this verification case, the background gas and the diffusive species are identical. The kinematic viscosity is given by:

$$\nu = 0.018829 \, \frac{\sqrt{T_0^3 \frac{2}{M}}}{P\sigma^2 \, \Omega_D(\epsilon/k)} = \frac{\alpha}{P} \tag{4.2}$$

where M the molar mass of the gas, σ and ϵ/k the Lennard–Jones parameters, and Ω_D the collision integral. The relevant thermophysical properties of the gas are summarized in Table 4.1.

Table 4.1: Gas properties

M (g $\cdot$ mol^{-1})	σ(Å)	ϵ/k (K)	α (kg $\cdot$ m $\cdot$ s^{-1})
29	3.617	97	2.087

As demonstrated in A.2, the analytical pressure profile is:

$$P(z)^2 = P_0^2 + \left(P_L^2 - P_0^2\right) \frac{z}{L}. \tag{4.3}$$

And the corresponding mass flux is:

$$\dot{m}'' = \frac{\bar{K}}{2\alpha L} \left(P_L^2 - P_0^2\right). \tag{4.4}$$

Figures 4.6, 4.5, and 4.7 present respectively the pressure distribution, the kinematic viscosity profile, and the mass flux comparison between the analytical solution and the Gpyro simulation.

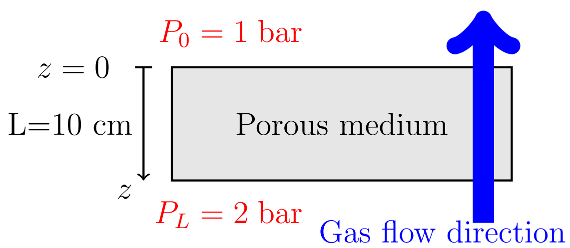


Figure 4.4: Configuration schematic

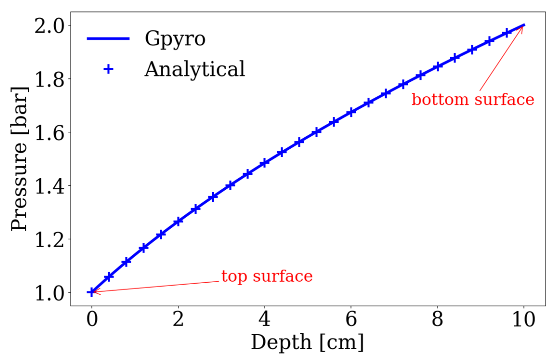


Figure 4.5: Pressure profile along the porous domain.

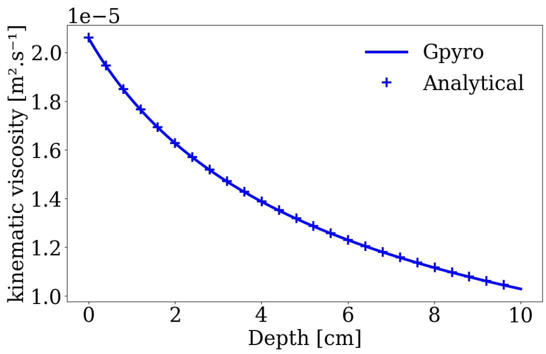


Figure 4.6: Kinematic viscosity profile.

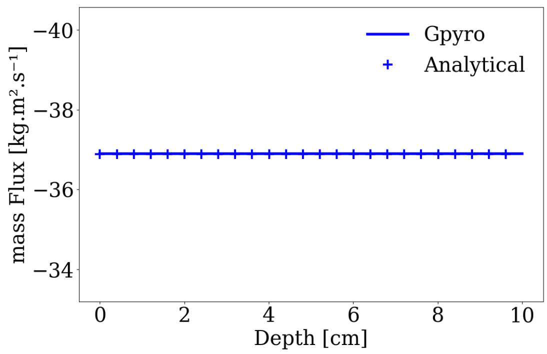


Figure 4.7: Mass flux along the domain.

4.3 Darcy Flow with Fixed Flux Boundary Conditions

This verification case examines the behaviour of the pressure solver in a porous material under steady one-dimensional gas flow with an imposed gas mass flux. A uniform porous layer of thickness $L = 10$ cm is considered (Figure 4.8). The temperature is fixed at $T_0 = 300$ K, and the permeability of the material is $\bar{K} = 5 \cdot 10^{-10}$ m^2. The background gas, gas properties and transport parameter are treated exactly as in Section 4.2.

At the top boundary, the pressure is imposed at 1 bar,while at the bottom boundary a mass flux is imposed:

$$\dot{m}''(L) = 10 \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}.$$

As shown in A.2, the analytical pressure profile is:

$$P(z) = \sqrt{P_0^2 + 2\frac{\alpha}{\bar{K}}\dot{m}_L'' \, z}. \tag{4.5}$$

The mass flux is spatially uniform and equal to the imposed boundary value:

$$\dot{m}'' = \dot{m}_L'' = 10 \text{ kg} \cdot \text{m}^{-2} \cdot \text{s}^{-1}. \tag{4.6}$$

Figures 4.10, 4.9, and 4.11 present respectively the pressure distribution, the kinematic viscosity profile, and the mass flux comparison between the analytical solution and the Gpyro simulation.

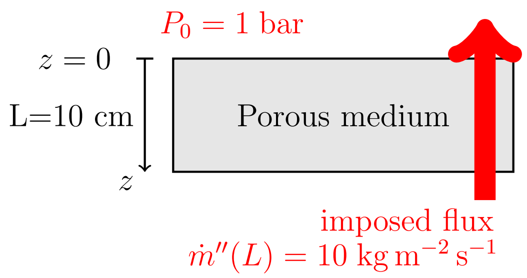


Figure 4.8: Configuration schematic.

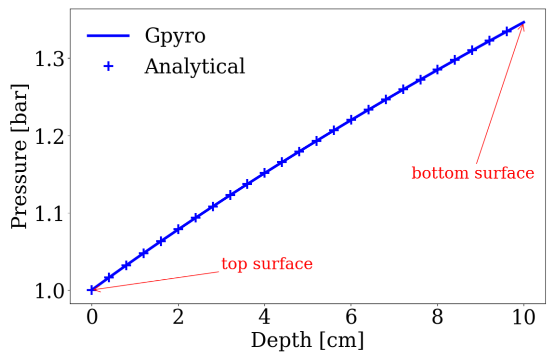


Figure 4.9: Pressure profile along the porous domain.

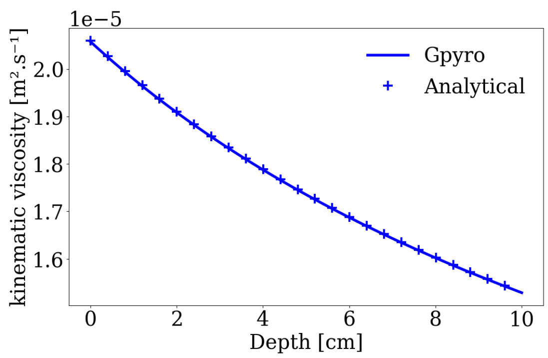


Figure 4.10: Kinematic viscosity profile.

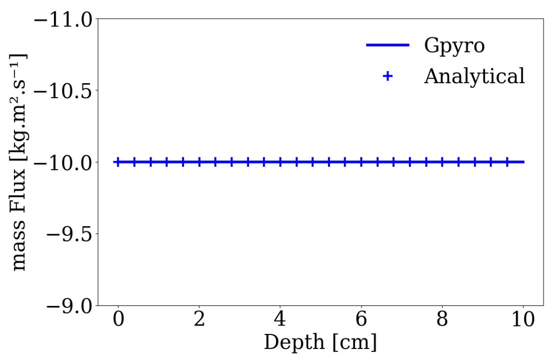


Figure 4.11: Mass flux along the domain.

4.4 Pure gas species diffusion

This verification case evaluates the gas species transport solver in the limit of *pure diffusion*. Numerical results are compared with an analytical solution for transient diffusion in a one-dimensional porous medium.

A one-dimensional porous layer of thickness $L = 1$ m is considered, with a uniform porosity $\overline{\psi} = 0.6$. The temperature and pressure are spatially uniform and remain constant at $T = 300$ K and $P = 1$ bar. Gas transport occurs solely by diffusion, with no advection and no volumetric source terms. The gaseous mixture consists of oxygen (O_2) and nitrogen (N_2), whose mass fractions are computed using the gas species diffusion solver. The external mass fractions are fixed to $Y^{\infty}_{O_2} = 0.22$ and $Y^{\infty}_{N_2} = 0.78$. Initially, the porous medium contains only nitrogen, such that $Y_{0,O_2} = 0$ and $Y_{0,N_2} = 1$ throughout the domain. Figure 4.12 illustrates the configuration.

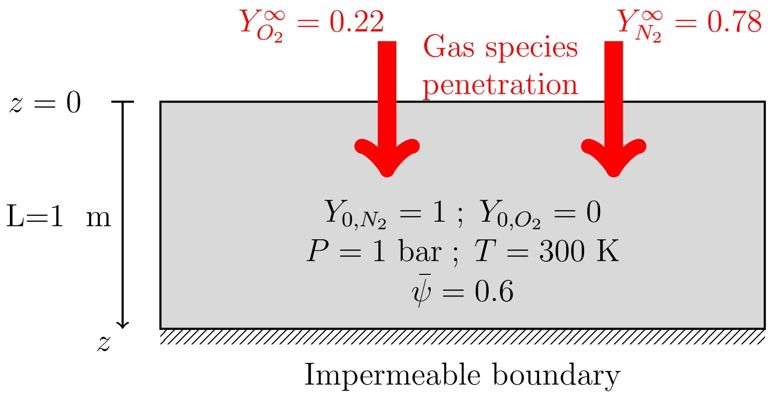


Figure 4.12: Configuration schematic

Under these assumptions, the transport equation for each gaseous species reduces to the diffusion equation

$$\frac{\partial Y_j}{\partial t} = D\,\nabla^2 Y_j \tag{4.7}$$

where Y_j is the species mass fraction and D is the effective diffusivity. Note that this equation is mathematically equivalent to the transient heat conduction equation.

The bottom boundary ($z = L$) is impermeable to diffusion:

$$\partial_z Y_j\big|_{z=L} = 0. \tag{4.8}$$

At the top boundary ($z = 0$), a convective boundary condition is imposed:

$$-\left[\overline{\psi}\,\rho_g\,D\,\partial_z Y_j\right]_{z=0} = h_m\left(Y^{\infty}_j - Y_j(0,t)\right) \tag{4.9}$$

where the mass transfer coefficient h_m is expressed in $\mathrm{kg\,m^{-2}\,s^{-1}}$.

Assuming a semi-infinite medium, the analytical solution of this problem is given by [1]:

$$Y_j(z,t) = Y_{0,j} - \left(Y_{0,j} - Y^{\infty}_j\right)\Phi(z,t) \tag{4.10}$$

where the dimensionless function $\Phi(z,t)$ is defined as:

$$\Phi(z,t) = \operatorname{erfc}\left(\frac{z}{2\sqrt{D\,t}}\right) - \exp\left(\alpha z + \alpha^2 D t\right)\operatorname{erfc}\left(\frac{z}{2\sqrt{D\,t}} + \alpha\sqrt{D\,t}\right) \tag{4.11}$$

and $\alpha = \frac{h_m}{\overline{\psi}\,\rho_g\,D}$.

The effective gas diffusivity is evaluated using the Chapman–Enskog model. The required gas molecular properties are reported in Table 4.2. Under the present thermodynamic conditions, the diffusivity is uniform and equal to $D = 0.20\ \mathrm{cm^2\,s^{-1}}$.

Table 4.2: Gas molecular properties

Species	M ($\mathrm{g\,mol^{-1}}$)	σ (Å)	ϵ/k (K)
O_2	32	3.433	113
N_2	28	3.681	91.5

Two values of the mass transfer coefficient are investigated: $h_m = 0.01\ \mathrm{kg\,m^{-2}\,s^{-1}}$ and $h_m = 10^{-4}\ \mathrm{kg\,m^{-2}\,s^{-1}}$.

For $h_m = 0.01\ \mathrm{kg\,m^{-2}\,s^{-1}}$, the surface resistance is negligible compared with diffusive transport inside the porous medium. In this regime, the boundary condition effectively reduces to a Dirichlet condition and the surface mass fractions rapidly converge towards the imposed external values Y_j^{∞}. This case therefore corresponds to fixed external concentrations.

Figures 4.13 and 4.14 present the evolution of the O_2 and N_2 mass fraction profiles for both values of h_m. Numerical results are compared with the analytical solution at several times, demonstrating excellent agreement.

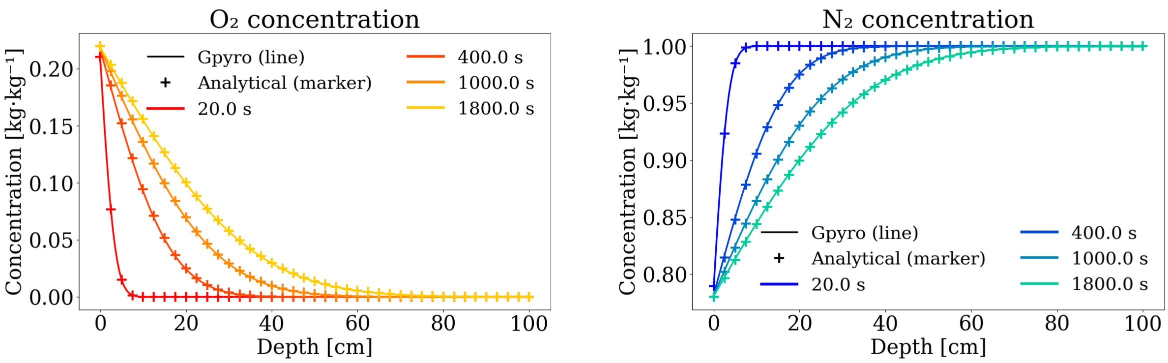


Figure 4.13: Species diffusion with $h_m = 0.01\ \mathrm{kg\,m^{-2}\,s^{-1}}$.

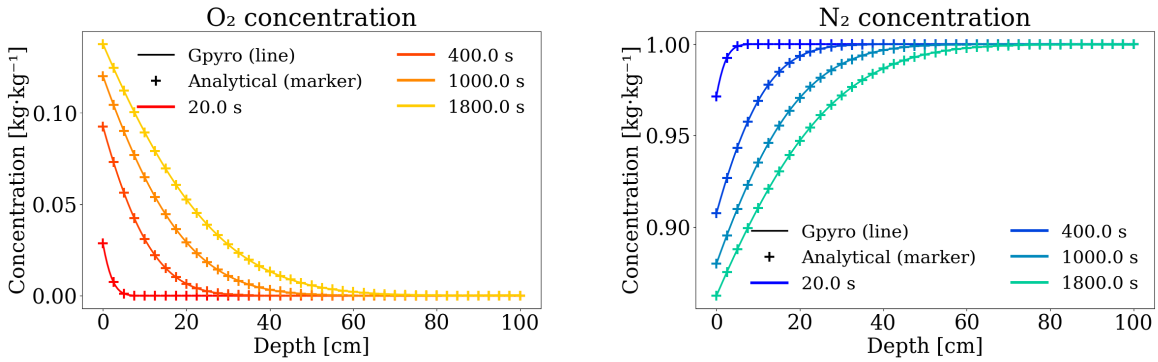


Figure 4.14: Species diffusion with $h_m = 10^{-4}\ \mathrm{kg\,m^{-2}\,s^{-1}}$.

4.5 Gas species advection

A one-dimensional porous layer of thickness $L = 10$ cm is considered, with a uniform porosity $\overline{\psi} = 0.6$. Gas transport occurs solely by advection, with no diffusion and no volumetric source terms. The gaseous mixture consists of oxygen (O_2) and nitrogen (N_2) and has a density $\rho_g = 1.3$ kg/m^3. The external mass flux is fixed to $\dot{m}'' = 10$ g.m^{-2}.s^{-1}. The entering gas mass fraction is $Y_{O_2}^{\infty} = 0.22$ and $Y_{N_2}^{\infty} = 0.78$. Initially, the porous medium contains only nitrogen, such that $Y_{0,O_2} = 0$ and $Y_{0,N_2} = 1$ throughout the domain.

Under these assumptions, the transport equation for each gaseous species j reduces to the advection equation:

$$\overline{\psi}\rho_g \frac{\partial Y_j}{\partial t} + \dot{m}'' \cdot \nabla Y_j = 0 \tag{4.12}$$

The top boundary ($z = 0$) is impermeable to diffusion:

$$\partial_z Y_j|_{z=0} = 0. \tag{4.13}$$

At the bottom boundary ($z = L$), a Dirichlet boundary condition is imposed:

$$Y_j(L, t) = Y_j^{\infty}. \tag{4.14}$$

This equation admits an analytical solution corresponding to the translation of a step profile at a constant velocity $v = \dot{m}''/\overline{\psi}\rho_g$. The analytical solution for the species mass fraction can therefore be written as

$$Y_j(z,t) = \begin{cases} Y_j^{\infty}, & L - z < vt \\ Y_{0,j}, & L - z \geq vt \end{cases} \tag{4.15}$$

The numerical simulations are performed using a uniform spatial discretization with $n_z = 500$ cells, corresponding to a grid spacing of $\Delta z = 0.2$ mm. A fixed time step of $\Delta t = 0.01$ s is used.

Figures 4.15 and 4.15 present the evolution of the O_2 and N_2 mass fraction profiles. Numerical results are compared with the analytical solution at several times.

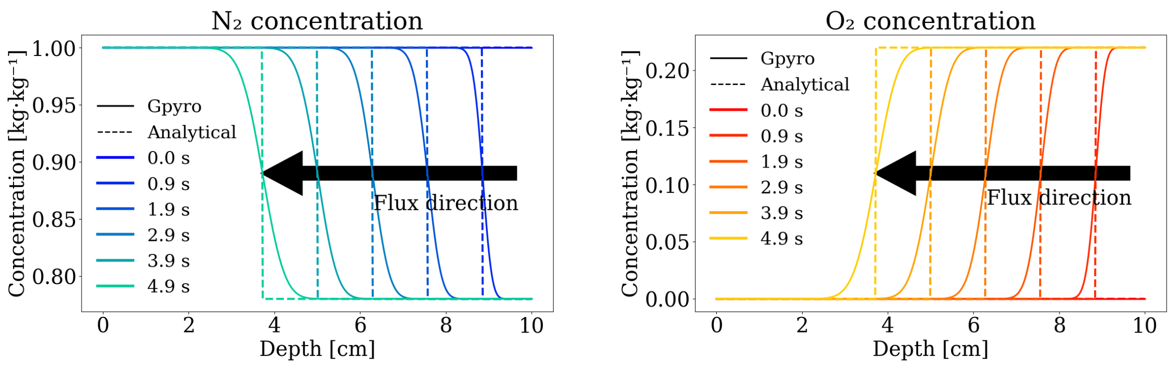


Figure 4.15: Species advection with 10 g/m^2/s.

The numerical solution accurately captures the propagation velocity of the front, as predicted by the analytical solution. However, the numerical solution exhibits a progressive smoothing of the front compared to the analytical step profile. Such diffusive behavior is due to the first-order spatial discretization of the advection scheme. This numerical diffusion is expected and decreases with smaller spatial steps. Users should be aware of this behavior.

4.6 Gas–solid heat exchange

This verification case evaluates the capability of GPYRO to correctly model heat exchange between a solid phase and a flowing gas under the assumption of local thermal equilibrium.

A one-dimensional slab of thickness $L = 10$ cm is considered. The solid is heated from the top surface by a constant external heat flux of 50 $\mathrm{kW\,m^{-2}}$. Simultaneously, a gas mass flux of $\dot{m}''_{\mathrm{gas}} = 0.1\ \mathrm{kg\,m^{-2}\,s^{-1}}$ enters the domain from the bottom boundary at an initial temperature of 300 K and flows upward through the porous medium.

At the inlet, the gas temperature is instantaneously equilibrated with the local solid temperature. Throughout the domain, gas and solid phases are assumed to remain in local thermal equilibrium, such that the gas and solid temperatures are equal. Under this assumption, in GPYRO the volumetric heat exchange term between the gas and the solid is expressed as:

$$\dot{Q}'''_{s-g} = c_{p,g}\,\dot{m}''_{\mathrm{gas}}\,\frac{\partial T}{\partial z} \tag{4.16}$$

where $c_{p,g} = 1000\ \mathrm{J\,kg^{-1}\,K^{-1}}$ is the gas specific heat capacity and T denotes the common gas–solid temperature.

The heat exchange term computed by GPYRO is compared against a semi-analytical reference obtained from Eq. (4.16), using the temperature gradient predicted by the numerical solution. The comparison is shown in Figure 4.16.

In addition, a second simulation is performed in which gas–solid heat exchange is disabled. This case serves as a reference to isolate the influence of the gas–solid coupling on the temperature field. The resulting temperature profiles for both configurations are compared in Figure 4.17, highlighting the impact of the gas flow on heat transport within the solid.

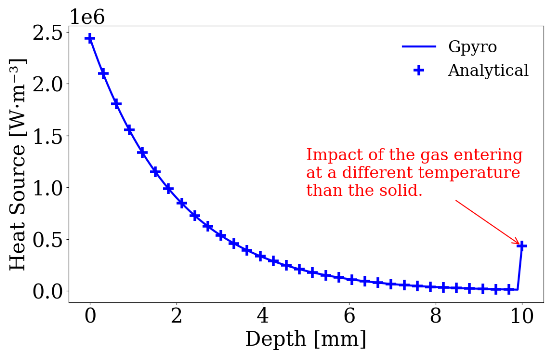


Figure 4.16: Volumetric gas–solid heat exchange term

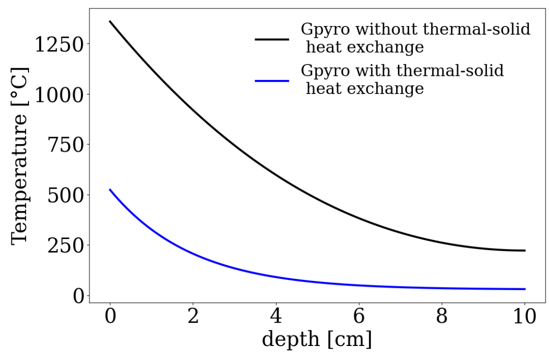


Figure 4.17: Temperature profiles with and without gas–solid exchange

Chapter 5

Multiphysics Verification

5.1 Cone Calorimeter Configuration

This verification case aims to assess the ability of Gpyro to reproduce a standard one-dimensional pyrolysis scenario representative of cone calorimeter tests, and to compare the results with those from the codes Thermakin and FDS, as detailed in the study [6].

A fictitious but physically coherent material, referred to as **Material A**, is used. Two independent first-order chemical reactions are modeled using the kinetic parameters detailed in Table 5.1, and the thermophysical properties of all solid species are provided in Table 5.2.

The physical configuration consists of a 6 mm thick layer of Material A placed atop a 12 mm Kaowool insulation layer. A constant external heat flux of 50 kW/m^2 is applied to the top surface. Convective heat transfer is imposed with a heat transfer coefficient of $h_c =$ 8.2 W/m^2·K and an ambient temperature of 25 °C. The bottom surface exchanges heat with the environment solely via thermal radiation. Figure 5.1 illustrates the configuration.

To ensure equivalence between the codes, the following modeling assumptions are adopted:

- Gas-phase contributions to energy conservation are neglected.
- Gas release is assumed to be instantaneous (no gas transport).
- Reaction enthalpies are treated as constant.
- No material deformation is modeled (densities chosen to preserve volume).

A uniform spatial discretization of 0.02 mm and a time step of 0.05 s are used. Numerical convergence is achived under these settings.

Table 5.1: Chemical kinetics parameters

Reaction	A_k (s^{-1})	E_k (kJ/mol)	n_k (-)	ΔH_k (J/kg)	θ_k (kg/kg)
A $\rightarrow \theta_1$ B + $(1-\theta_1)$ Gas$_1$	9.5×10^{20}	249	1	1.7×10^5	0.44
B $\rightarrow \theta_2$ C + $(1-\theta_2)$ Gas$_2$	5.5×10^{11}	192	1	1.2×10^6	0.47

Table 5.2: Thermophysical properties of solid species

Material	k (W/m·K)	c_p (J/kg·K)	ρ (kg/m^3)	α (m^{-1})	ε (-)
Material A	$0.17 + 3.10^{-4}T$	1550	1430	∞	1.0
Material B	$0.24 + 8.10^{-4}T$	1720	629.2	∞	1.0
Material C	$0.36 + 18.10^{-4}T$	780	265.7	∞	1.0
Kaowool	0.058	1070	256	∞	0.95

The results obtained using Gpyro are compared to those from Thermakin and FDS. Figure 5.2 presents the comparison of the Mass Loss Rate (MLR) profiles. Figure 5.3 illustrates the evolution of temperature at various depths, while Figure 5.4 shows the time-dependent evolution of species concentrations at a depth of 3 mm.

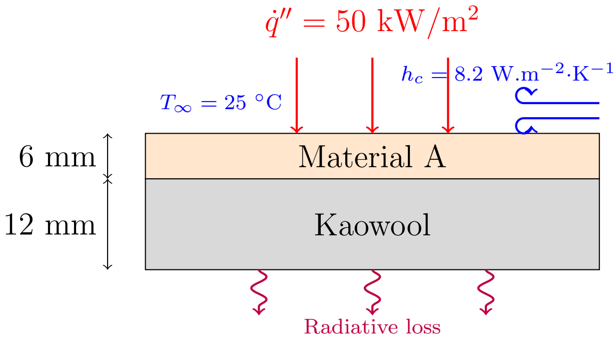


Figure 5.1: Configuration schematic

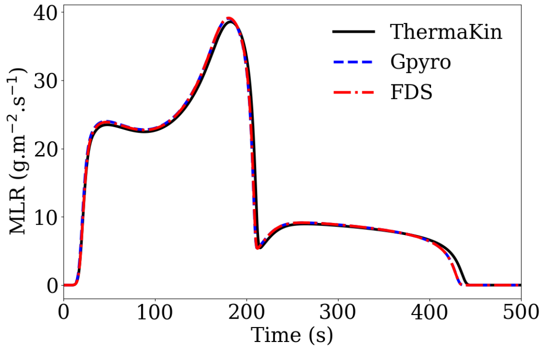


Figure 5.2: Mass Loss Rate (MLR) comparison

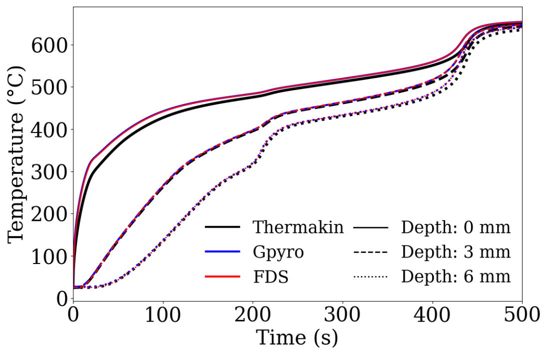


Figure 5.3: Temperature evolution at different depths

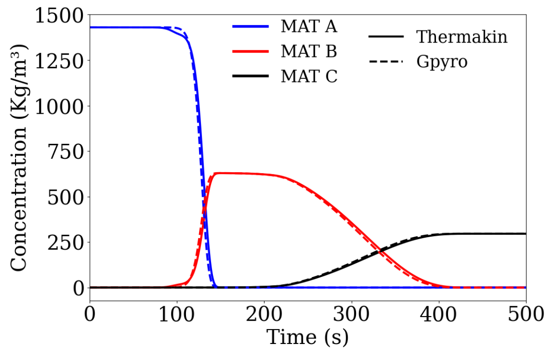


Figure 5.4: Species concentration evolution at 3 mm depth

5.2 Extension to 2D and 3D Configurations

To further evaluate the robustness and consistency of the solver, the reference one-dimensional cone calorimeter configuration (See Section 5.1) is extended to two- and three-dimensional domains. The domain is expanded to 10 cm $\times$ 10 cm in the x and y directions, while maintaining a vertical thickness of 18 mm identical to the 1D case.

The computational mesh is composed of 20 uniform cells in both the x and y directions, and 180 cells in the z direction. This corresponds to a spatial resolution of $\Delta x = \Delta y =$ 5 mm and $\Delta z = 0.1$ mm, which is 50 times finer along the vertical axis. A constant time step of $\Delta t = 0.5$ s is employed throughout the simulation.

In order to ensure theoretical equivalence with the one-dimensional configuration, adiabatic boundary conditions are imposed on all lateral surfaces (in the x and y directions). Under these assumptions, the 2D and 3D configurations are expected to reproduce the same thermal and chemical behavior as the 1D case.

The comparison of the Mass Loss Rate (MLR) obtained from the 1D, 2D, and 3D simulations is presented in Figure 5.5. The consistency between the results confirms the proper extension of the solver to multidimensional domains without introducing numerical bias.

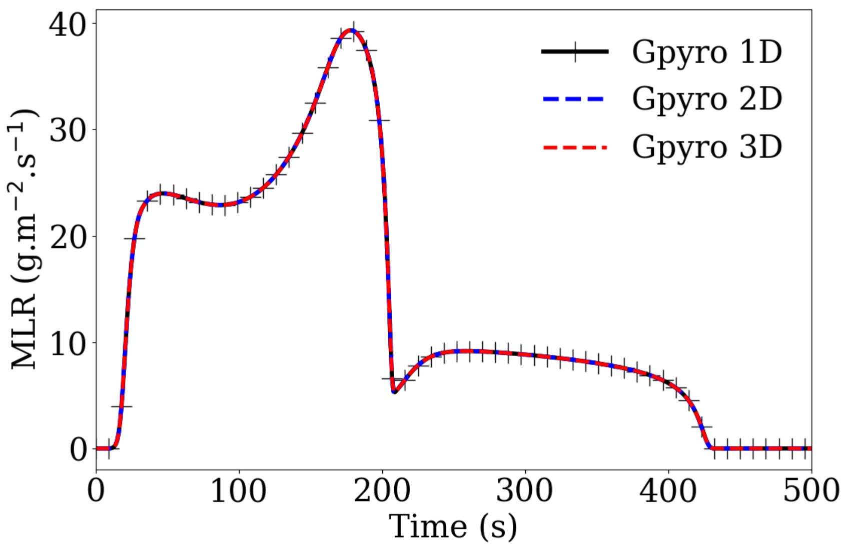


Figure 5.5: Comparison of Mass Loss Rate (MLR) between 1D, 2D, and 3D configurations

5.3 3D Cone Calorimeter with Lateral Convection

This case extends the reference cone calorimeter configuration (See Section 5.1) into three dimensions by introducing natural convection on the lateral x and y surfaces, in addition to the top-applied heat flux. This setup induces multidimensional effects in the degradation front propagation, resulting in a more realistic physical behavior.

The domain consists of a 6 cm thick slab of **Material A** atop a 12 cm Kaowool insulation layer, with lateral dimensions of 10 cm $\times$ 10 cm. A vertical mid-plane at $x = 5$ cm (shown in gray in Figure 5.6) is used to visualize the pyrolysis front using Smokeview. Figure 5.7 shows the reaction rate distribution in this plane at various time steps.

The computational grid uses uniform spacing: $\Delta z = 0.1$ mm to resolve vertical gradients, and $\Delta x = \Delta y = 5$ mm laterally. A fixed time step of $\Delta t = 0.5$ s is applied.

Figure 5.7 illustrates the in-depth propagation of the degradation front and the influence of convective heat losses along the domain boundaries.

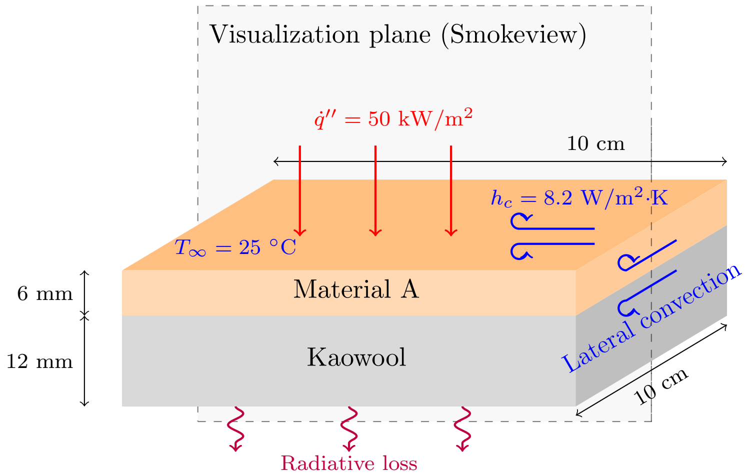


Figure 5.6: Three-dimensional configuration.

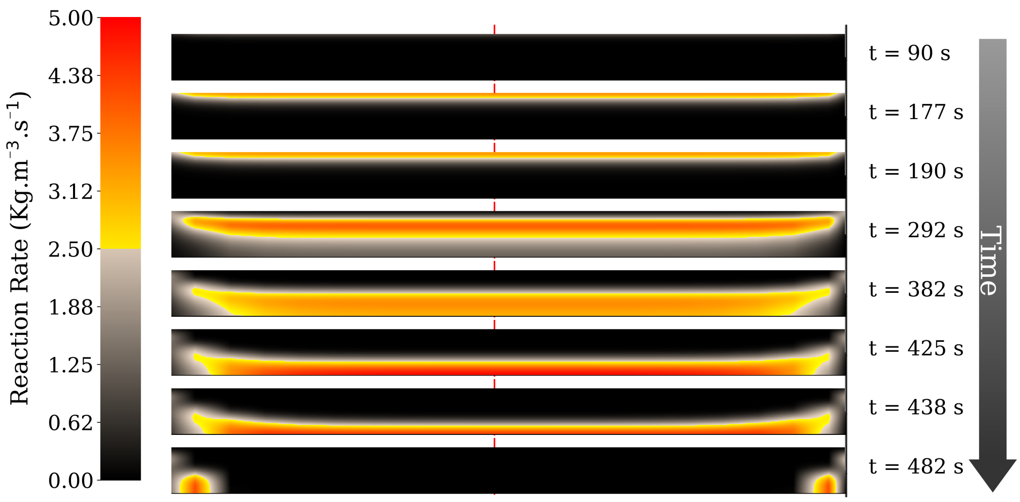


Figure 5.7: Reaction rate distribution in the mid-height plane YZ at different times.

Chapter 6

Numerical Performance

6.1 Convergence Of The Species Solver

This verification case evaluates the temporal convergence of the species conservation solver and determines its temporal order of accuracy. Only a temporal study is required because the species conservation equation contains no spatial derivative term and reduces to a local balance.

The configuration corresponds to the simple TGA verification case (see Section 3.3). Several simulations were performed with time steps ranging from 100 s down to 10^{-2} s. For large time steps (e.g. 100 s), the species correction step of the solver is expected to be activated to ensure that the overall species destruction balance remains consistent within each computational cell.

The evolution of the mass loss rate (MLR) obtained with the different time steps is shown in Figure 6.1. The numerical error is quantified using the L^2 norm:

$$L^2 = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\phi_i^{\text{num}} - \phi_i^{\text{ref}}\right)^2} \tag{6.1}$$

where ϕ_i^{num} and ϕ_i^{ref} denote the numerical and reference solutions at time index i, and N is the total number of compared points. The reference solution corresponds to the simulation performed with a time step of 10^{-3} s. The resulting L^2 error as a function of the time step is presented in Figure 6.2. A linear slope equal to 1 is obtained in the log–log error plot, which is expected since the species solver is first order accurate in time.

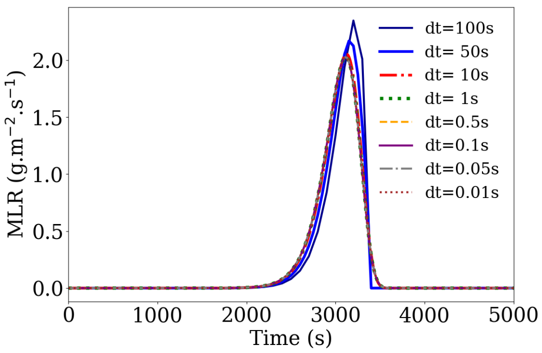


Figure 6.1: Mass Loss Rate (MLR) obtained for several time steps.

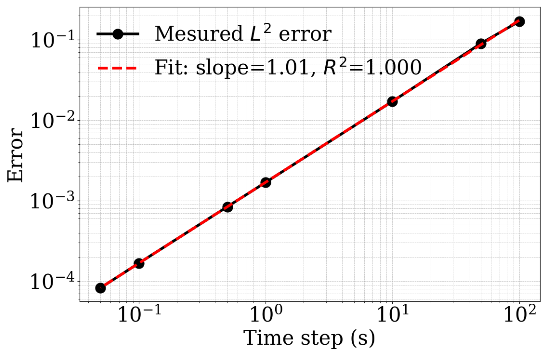


Figure 6.2: L^2 error of the species solver as a function of the time step.

6.2 Convergence Of The Thermal Solver

This verification case evaluates the temporal and spatial convergence of the one-dimensional thermal solver. The configuration corresponds to the case presented in Section 1.7. A slab of thickness 4 cm is considered, with a prescribed heat flux applied at the upper boundary.

Temporal convergence

The temporal convergence is examined by varying the time step from 0.01 s to 100 s, while maintaining a uniform spatial discretization of 0.1 mm. Although the implicit formulation ensures stability for all tested time steps, significant deviations from the analytical solution are expected when coarse temporal resolutions are used.

The temperature at a depth of 8 mm is monitored over time. Figure 6.3 presents the computed temperature histories, and Figure 6.4 shows the associated L^2 error (Eq. 6.1) relative to the analytical solution. A linear slope equal to 1 is obtained in the log–log error plot, which is expected since the thermal solver is first order accurate in time.

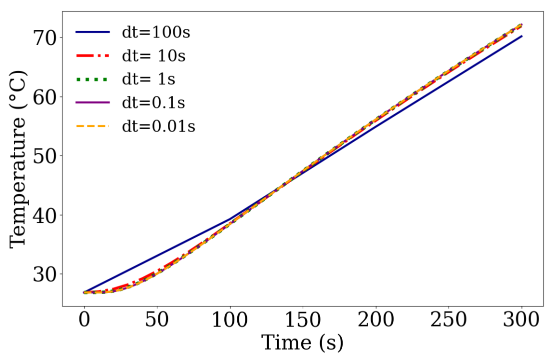


Figure 6.3: Temperature at 8 mm depth for various time steps.

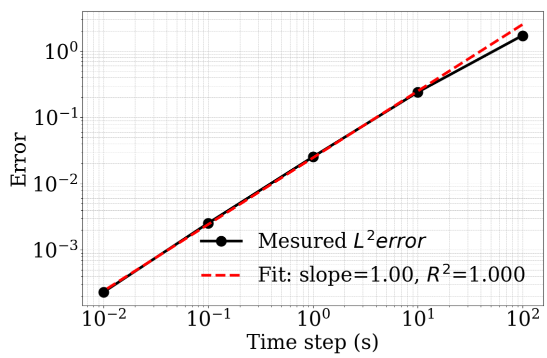


Figure 6.4: L^2 temporal error as a function of time step.

Spatial convergence

The spatial convergence is evaluated by varying the grid spacing from 10 mm (five cells through the slab) down to 0.001 mm. All simulations are run using a fixed time step of 0.01 s. Unconditional stability is preserved regardless of the grid refinement, although coarse meshes are expected to deviate from the analytical solution.

The temperature at 8 mm depth is again used to compute the L^2 error. Figure 6.5 presents the computed temperature histories. Figure 6.6 presents the spatial error as a function of grid spacing. A linear slope equal to 2 is obtained in the log–log error plot, which is expected since the thermal solver is second order accurate in space.

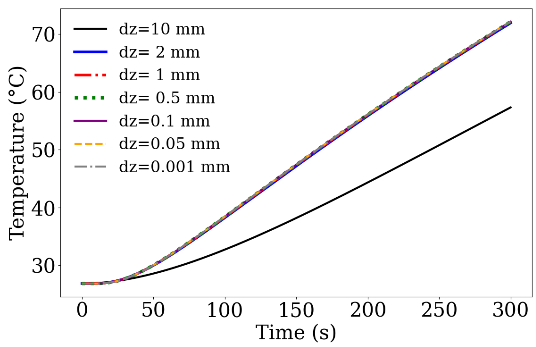


Figure 6.5: Temperature at 8 mm depth for various time steps.

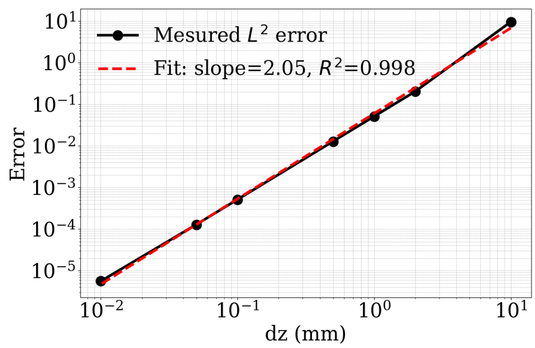


Figure 6.6: L^2 spatial error as a function of grid spacing.

Computational cost

The computational cost of the simulations is reported in Figures 6.7 and 6.8. A proportional increase in CPU time is expected as the spatial or temporal resolution is refined. For coarse discretizations, however, initialization and overhead operations may dominate the total runtime, and meaningful scaling behaviour is observed only once the discretization becomes sufficiently fine.

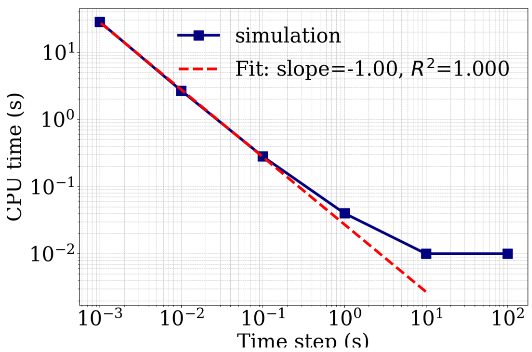


Figure 6.7: CPU time for temporal convergence simulations.

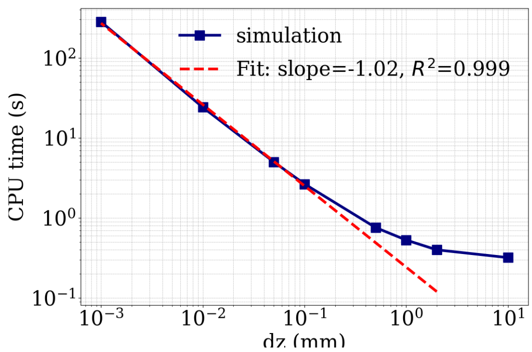


Figure 6.8: CPU time for spatial convergence simulations.

6.3 Convergence of the Cone Calorimeter Solver

This verification case evaluates the temporal and spatial convergence of the coupled thermal and pyrolysis solvers in a cone calorimeter configuration. The reference simulation corresponds to the case presented in Section 5.1. The mass loss rate (MLR) is used as the primary quantity of interest to assess numerical convergence.

Temporal convergence

The temporal convergence is examined by varying the time step from 10 s to 0.0005 s, while maintaining a uniform spatial discretization of 0.1 mm.

Figure 6.9 presents the mass loss rate histories obtained for the different temporal discretizations. Figure 6.10 presents the corresponding error as a function of time step size. The error is computed using the L^2 norm by comparing each solution with the reference solution obtained using the smallest time step (0.0005 s). A linear slope equal to 1 is obtained in the log–log error plot, which is consistent with the first-order temporal accuracy of the numerical scheme.

In addition, For time steps smaller than 1 s, the solution exhibits convergence. For larger time steps, deviations appear due to temporal discretization errors. Nevertheless, because the one-dimensional thermal solver is implicit and therefore unconditionally stable, these deviations remain bounded and manifest as oscillations rather than instabilities. Moreover, the species conservation equation is implemented in conservative form. As a result, even with large time steps, the total mass consumed remains nearly unchanged.

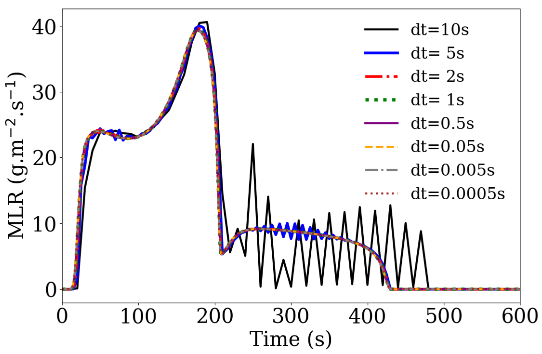


Figure 6.9: Mass loss rate for various time steps.

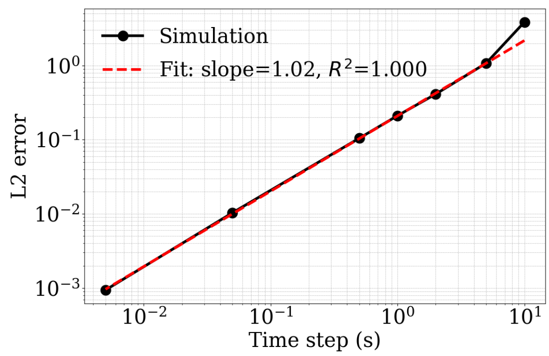


Figure 6.10: Temporal error as a function of time step size.

Spatial convergence

The spatial convergence is evaluated by varying the grid spacing from 1 mm down to 10^{-4} mm. All simulations are run using a fixed time step of 0.05 s.

Figure 6.11 presents the mass loss rate histories obtained for the different spatial discretizations. Figure 6.12 presents the corresponding L^2 error as a function of grid spacing. The mass loss rate obtained with the finest mesh is used as the reference solution for computing the error.

A linear slope close to 1 is obtained in the log–log error plot, indicating first-order spatial convergence for the complete coupled pyrolysis problem. This behavior is consistent with the first-order spatial accuracy of the mass conservation solver, which dominates the overall error of the coupled system.

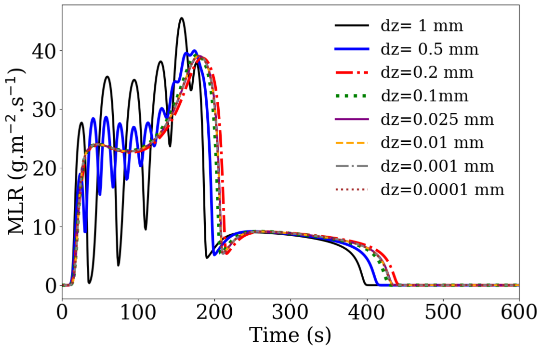


Figure 6.11: Mass loss rate for various grid spacings.

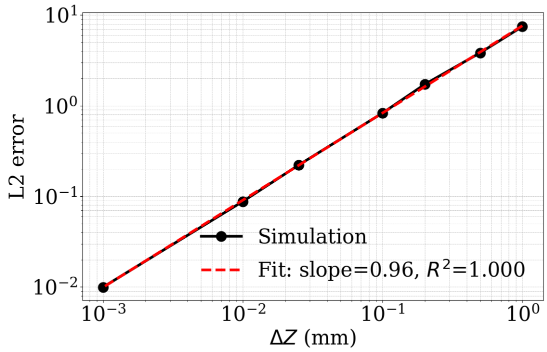


Figure 6.12: Spatial error as a function of grid spacing.

Computational cost

The computational cost associated with the temporal and spatial convergence studies is reported in Figures 6.13 and 6.14. As expected, the computational time increases linearly whith time and spatial discretisation.

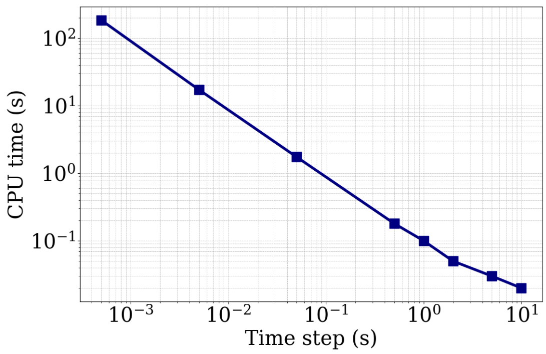


Figure 6.13: CPU time for temporal convergence simulations.

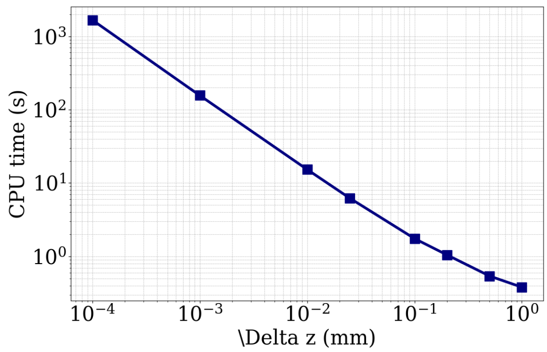


Figure 6.14: CPU time for spatial convergence simulations.

6.4 Verification of OpenMP Parallelization

This verification case reproduces the reference cone calorimeter setup (Section 5.1). The simulation was tested with different numbers of CPUs to evaluate parallel performance and robustness.

The chosen mesh contains 5000 cells, with a fixed time step of $\Delta t = 0.5\,\mathrm{s}$ and a total simulation time of $200\,\mathrm{s}$.

Parallel efficiency is analyzed using the *speedup*, defined as the ratio of the CPU time on a single processor to the CPU time on N processors:

$$S(N) = \frac{T(1)}{T(N)} \tag{6.2}$$

where $T(N)$ is the execution time using N processors.

Amdahl's law is used to estimate the parallelizable fraction f of the computation, which represents the portion of the program that can be executed in parallel. The remaining fraction $1 - f$ corresponds to the inherently sequential part. This relationship is expressed as:

$$S(N) = \frac{1}{(1-f) + \frac{f}{N}} \tag{6.3}$$

where N is the number of processors.

If the estimated parallelizable fraction f exceeds approximately 65%, the parallelization is considered effective, providing significant performance gains.

Additionally, the mass loss rate (MLR) output is monitored to ensure result consistency. Incorrect handling of shared or private variables during parallelization can cause bugs and result in differing outputs depending on the CPU count.

It is generally recommended not to assign fewer than 1000 cells per CPU, as below this threshold, the benefits of parallelization diminish relative to CPU resources allocated. Communication overhead between CPUs may outweigh computation time, potentially making parallelization more costly than beneficial.

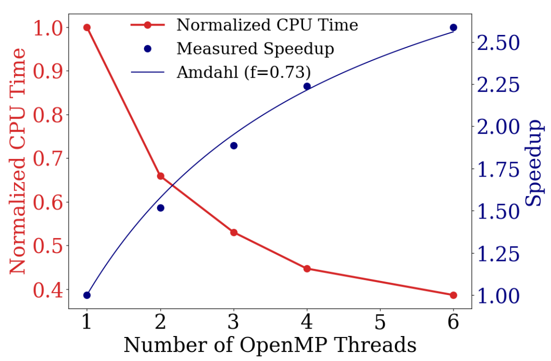


Figure 6.15: Speedup with fitting by Amdahl's law and CPU time.

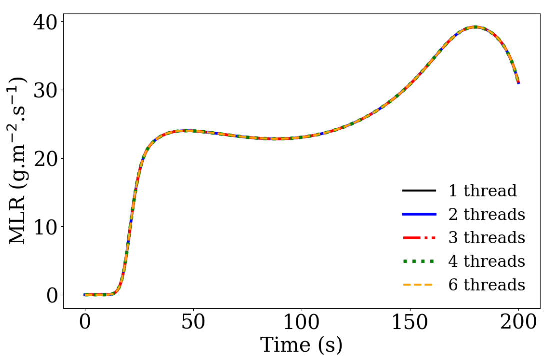


Figure 6.16: MLR for different numbers of CPUs.

Chapter 7

Validation

7.1 PMMA Validation Case (MaCFP Benchmark)

This validation case reproduces the MaCFP benchmark experiment on black **poly(methyl methacrylate) (PMMA)** performed in the NIST Gasification Apparatus under an external radiant heat flux of **50** kW/m^2 [7]. The sample consists of a **5.9** mm thick PMMA slab placed on a **12** mm Kaowool insulation layer, with the back surface fully insulated (adiabatic). PMMA is modeled as a non-charring material, decomposing into gas through a single first-order reaction. The kinetic parameters are summarized in Table 7.1, while the thermophysical properties of PMMA and Kaowool are given in Table 7.2. The simulation uses a uniform spatial discretization of $\Delta z = 0.025$ mm and a time step of $\Delta t = 0.05$ s. The results of the simulation are presented in Figure 7.1.

Table 7.1: Chemical kinetics parameters for PMMA

Reaction	A_k (s^{-1})	E_k (kJ/mol)	n_k (-)	ΔH_k (J/kg)
PMMA $\rightarrow$ Gas	1.35×10^{11}	164	1	8.17×10^5

Table 7.2: Thermophysical properties of PMMA and Kaowool

Material	k (W/m·K)	c_p (J/kg·K)	ρ (kg/m^3)	α (m^{-1})	ε (-)
PMMA ($T \leq 394$K)	0.16	$-1390 + 8.33T$	1210	2870	0.96
PMMA ($T > 394$K)	$0.3 - 4.2 \times 10^{-4}T$	$851 + 3.07T$	1210	2870	0.96
Kaowool	0.058	1070	256	∞	0.95

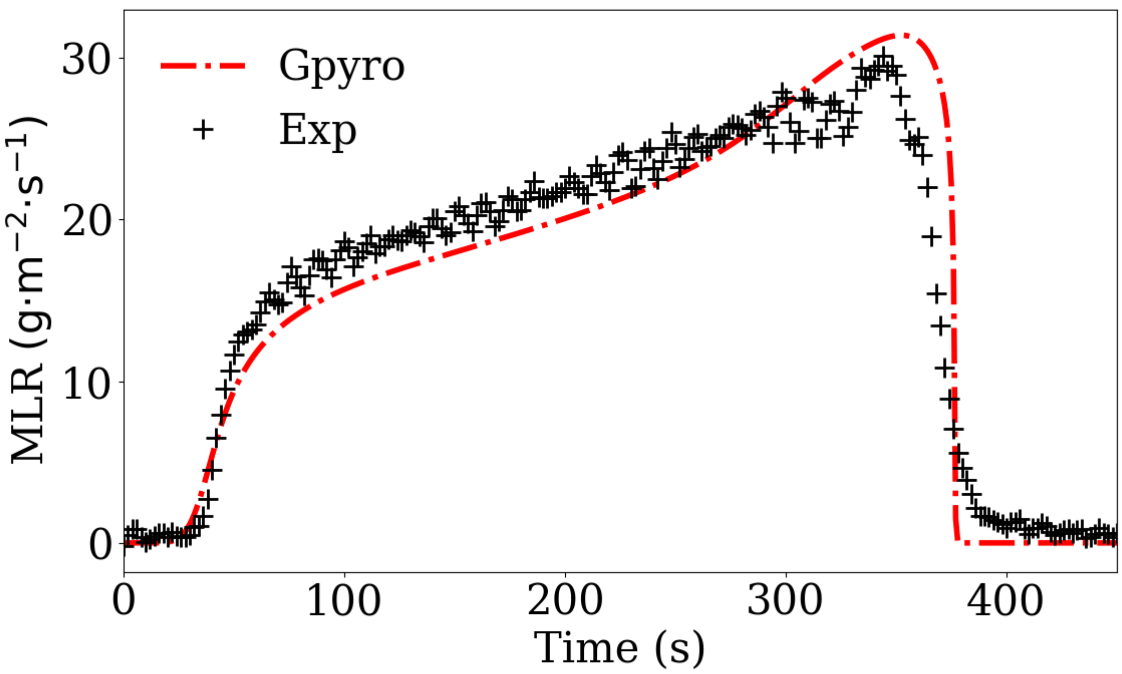


Figure 7.1: Simulation of the Mass Loss Rate (MLR) for PMMA in the NIST Gasification Apparatus

7.2 Smoldering in Polyurethane Foam

This case reproduces an experiment on the forward propagation of a smoldering front through a polyurethane foam cylinder conducted under microgravity conditions aboard the NASA Space Shuttle [8, 9, 10]. This configuration has been used as a one-dimensional validation case for Gpyro in the PhD thesis of Lautenberger [11, 12] and is also documented in the smoldering combustion workbook of Rein [13]. The same experiment has subsequently been simulated in two dimensions [14].

7.2.1 Experimental Configuration

The sample consists of a cylindrical polyurethane foam specimen with a diameter of 12 cm and a length of $L = 14$ cm. Ignition is initiated at the bottom face using an external igniter. Air is injected at the igniter side, such that the smoldering front propagates in the same direction as the imposed airflow. A schematic overview of the experimental configuration is provided in Figure 7.2.

The superficial air velocity is maintained at 0.1 $\mathrm{mm\,s^{-1}}$ during the first 400 s. It is then increased linearly to 5 $\mathrm{mm\,s^{-1}}$ between 400 s and 410 s and remains constant for the remaining 600 s of the experiment.

Temperatures are measured using centerline thermocouples located at eight axial positions. These temperature histories are compared with numerical predictions, from which the smolder propagation velocity is inferred and compared with experimental data.

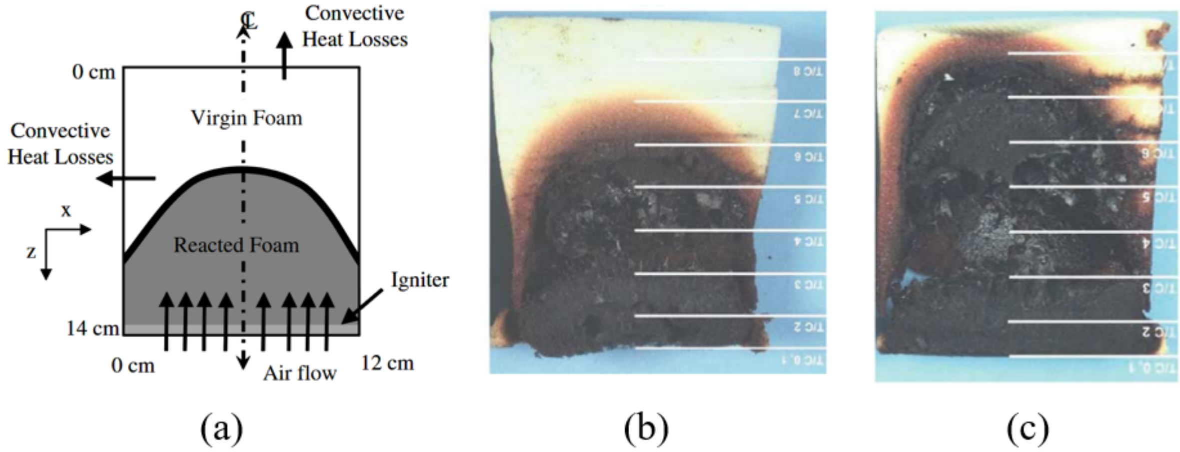


Figure 7.2: (a) Schematic representation of the experimental setup [14]. (b) and (c) images of the polyurethane foam sample after the experiment [9].

7.2.2 Numerical Model

The experiment is simulated using GPYRO with gas transport solvers activated and by modeling the oxidation. The simulation is performed in one dimension. A time step of $\Delta t = 0.2$ s is used, together with a spatial grid resolution of 0.25 mm. Numerical convergence has been verified with respect to both the time step and the spatial resolution.

Chemical kinetics: The smoldering process is modeled using a seven-step kinetic mechanism [14]. Solid species are denoted by the subscript (s) and gaseous species by

(g). The reactions are written using solid-phase stoichiometric coefficients θ_i and oxygen coefficients $\theta_{i,\mathrm{O_2}}$:

$$\text{foam (s)} \rightarrow \theta_1\, \beta\text{-foam (s)} + (1-\theta_1)\, \text{T-pyrolysate (g)} \tag{7.1}$$
$$\beta\text{-foam (s)} \rightarrow \theta_2\, \text{T-char (s)} + (1-\theta_2)\, \text{T-pyrolysate (g)} \tag{7.2}$$
$$\text{T-char (s)} \rightarrow \text{T-pyrolysate (g)} \tag{7.3}$$
$$\text{foam (s)} + \theta_{4,\mathrm{O_2}}\, \mathrm{O_2} \rightarrow \theta_4\, \text{char (s)} + (1-\theta_4+\theta_{4,\mathrm{O_2}})\, \mathrm{O_2}\text{-pyrolysate (g)} \tag{7.4}$$
$$\beta\text{-foam (s)} + \theta_{5,\mathrm{O_2}}\, \mathrm{O_2} \rightarrow \theta_5\, \text{char (s)} + (1-\theta_5+\theta_{5,\mathrm{O_2}})\, \mathrm{O_2}\text{-pyrolysate (g)} \tag{7.5}$$
$$\text{char (s)} + \theta_{6,\mathrm{O_2}}\, \mathrm{O_2} \rightarrow \theta_6\, \alpha\text{-char (s)} + (1-\theta_6+\theta_{6,\mathrm{O_2}})\, \mathrm{O_2}\text{-pyrolysate (g)} \tag{7.6}$$
$$\alpha\text{-char (s)} \rightarrow \text{products (g)} \tag{7.7}$$

Kinetic parameters: The Arrhenius parameters associated with reactions (7.1)–(7.7) are summarized in Table 7.3. The Arrhenius pre-exponential factor is denoted by Z, the activation energy by E, the reaction order by n, and the oxygen reaction order by $n_{\mathrm{O_2}}$. The reaction enthalpy is denoted by ΔH. The table also reports the solid stoichiometric coefficients θ_i and the oxygen coefficients $\theta_{i,\mathrm{O_2}}$.

Table 7.3: Kinetic parameters for the polyurethane foam smoldering reactions.

Reaction	Z ($\mathrm{s^{-1}}$)	E ($\mathrm{kJ\,mol^{-1}}$)	n	$n_{\mathrm{O_2}}$	θ	$\theta_{\mathrm{O_2}}$	ΔH ($\mathrm{J\,kg^{-1}}$)
(7.1)	3.16×10^{18}	227.0	0.80	0	0.68	0.0	4.0×10^{4}
(7.2)	1.29×10^{10}	146.5	1.25	0	0.061	0.0	7.5×10^{5}
(7.3)	8.24×10^{8}	173.3	0.92	0	0.0	0.0	-2.5×10^{6}
(7.4)	1.37×10^{15}	188.0	0.48	1	0.38	0.186	-1.5×10^{6}
(7.5)	1.00×10^{16}	200.0	0.52	1	0.56	0.176	-1.6×10^{6}
(7.6)	1.00×10^{15}	201.0	1.30	1	0.24	1.14	-2.5×10^{6}
(7.7)	4.25×10^{8}	153.0	1.61	1	0.0	1.5	-2.5×10^{6}

Condensed-phase properties: The condensed-phase properties are reported in Table 7.4. The temperature-dependent effective thermal conductivity is defined as $k_{\mathrm{eff}} = k_0(T/T_{\mathrm{ref}})^{n_k} + \gamma\sigma T^3$, where γ represents the pore radiation contribution and σ is the Stefan–Boltzmann constant. The quantities ρ and ρ_s denote the bulk density and the skeleton density, respectively. The heat capacity is defined as $c_p = c_0(T/T_{\mathrm{ref}})^{n_c}$. The emissivity is denoted by ϵ, and K is the permeability.

Table 7.4: Condensed-phase thermophysical properties.

Species	k_0	n_k	γ	ρ	ρ_s	c_0	n_c	ϵ	K
	$\mathrm{W\,m^{-1}\,K^{-1}}$	-	m	$\mathrm{kg\,m^{-3}}$	$\mathrm{kg\,m^{-3}}$	$\mathrm{J\,kg^{-1}\,K^{-1}}$	-	-	$\mathrm{m^2}$
Foam	0.05	1.6	0.001	26.5	900	1760	0.7	1.0	5.2×10^{-9}
β-foam	0.05	1.6	0.001	18.0	900	1760	0.7	1.0	1×10^{-8}
T-char	0.05	1.6	0.001	1.1	900	1760	0.7	1.0	3×10^{-8}
Char	0.05	1.6	0.001	10.0	900	1760	0.7	1.0	3×10^{-8}
α-char	0.05	1.6	0.001	2.4	900	1760	0.7	1.0	3×10^{-8}

Gas-phase properties: The gas-phase heat capacity is assumed constant, with $c_{p,g} = 1100\ \mathrm{J\,kg^{-1}\,K^{-1}}$. Molecular weights noted M and Lennard–Jones parameters used for diffusivity estimation are listed in Table 7.5.

Table 7.5: Gas-phase thermophysical properties.

Species	M ($\mathrm{g\,mol^{-1}}$)	σ (Å)	ϵ/k (K)
O_2	32	3.434	113
N_2	28	3.667	99.8
T-pyrolysate	44	3.996	190
O_2-pyrolysate	44	3.996	190
Product	44	3.996	190

Boundary conditions: At the bottom boundary ($z = L$), the experimentally measured igniter temperature is imposed as a thermal boundary condition. At the top boundary ($z = 0$), heat losses are modeled using a convective heat transfer coefficient of $10\ \mathrm{W\,m^{-2}\,K^{-1}}$ combined with surface reradiation. Additionally, as the simulation is one-dimensional, radial heat losses are accounted for using a volumetric heat loss coefficient of $25\ \mathrm{W\,m^{-3}\,K^{-1}}$.

For the pressure solver, a mass flux of $5.8\ \mathrm{kg\,m^{-2}\,s^{-1}}$ is imposed at the inlet ($z = L$), corresponding to an air velocity of $5\ \mathrm{mm\,s^{-1}}$ and an air density of $1.2\ \mathrm{kg\,m^{-3}}$. Atmospheric pressure (1 bar) is imposed at the outlet.

For gas species, inlet mass fractions are fixed to $Y_{\mathrm{O_2}} = 0.23$ and $Y_{\mathrm{N_2}} = 0.77$.At the outlet, the flow is assumed to be dominated by advection (high Peclet number), and a free-outlet condition is imposed, where the concentration gradient is negligible and the gas exits freely:

$$\frac{\partial Y_j}{\partial z} = 0 \tag{7.8}$$

7.2.3 Results

Temperature evolution. Figure 7.3 compares the predicted temperature histories at different axial locations with the experimental thermocouple measurements. The numerical results reproduce the main features of the experiment, including the arrival time of the thermal front and the order of magnitude of the peak temperatures. Overall, the agreement indicates that the thermal response of the foam is captured satisfactorily.

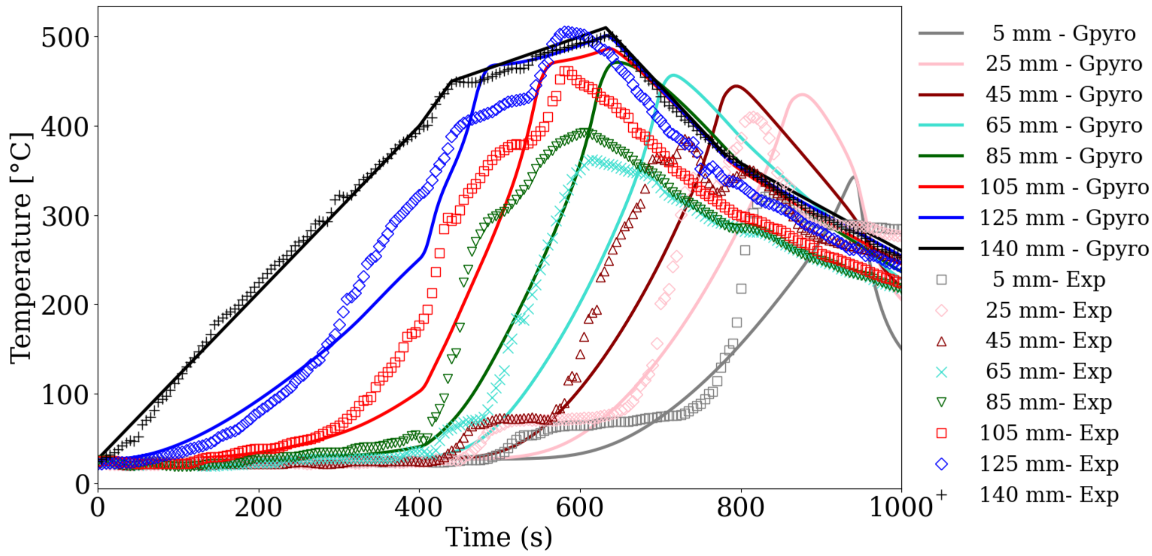
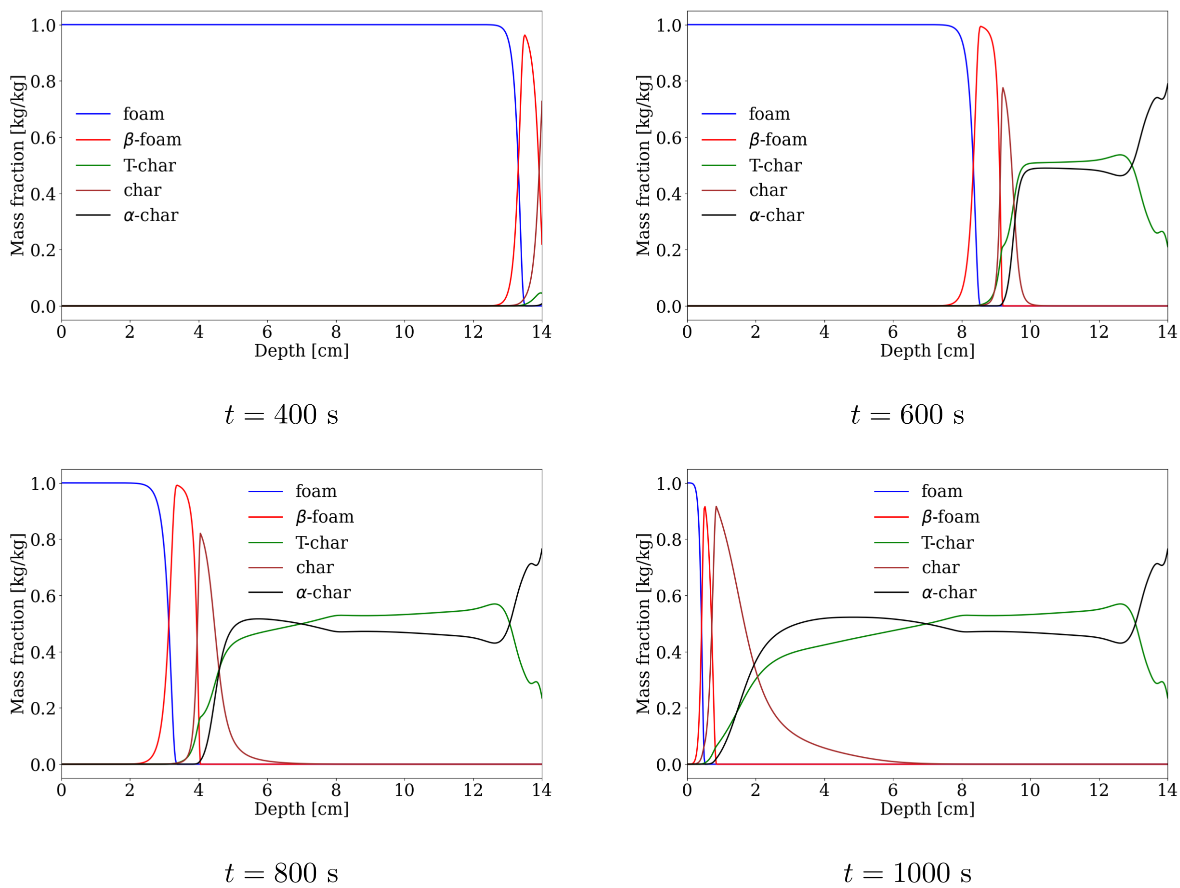


Figure 7.3: Comparison between predicted and measured temperature histories at different axial positions.

Solid-phase composition. The evolution of the solid-phase composition is shown in Figure 7.4 at four representative times: $t = 400$ s, corresponding to the early stage of degradation; $t = 600$ s and $t = 800$ s, during active smolder propagation; and $t = 1000$ s, near the end of the process. Degradation initiates with the formation of β-foam, which result from purely pyrolytic reactions (reaction 7.1). Slightly downstream, a second reaction zone develops, associated with oxidation reactions and the formation of char (reaction 7.4 and 7.5). After the passage of these fronts, the remaining solid residue is mainly composed of α-char and T-char, indicating that the smoldering reactions have largely completed.



$t = 400$ s

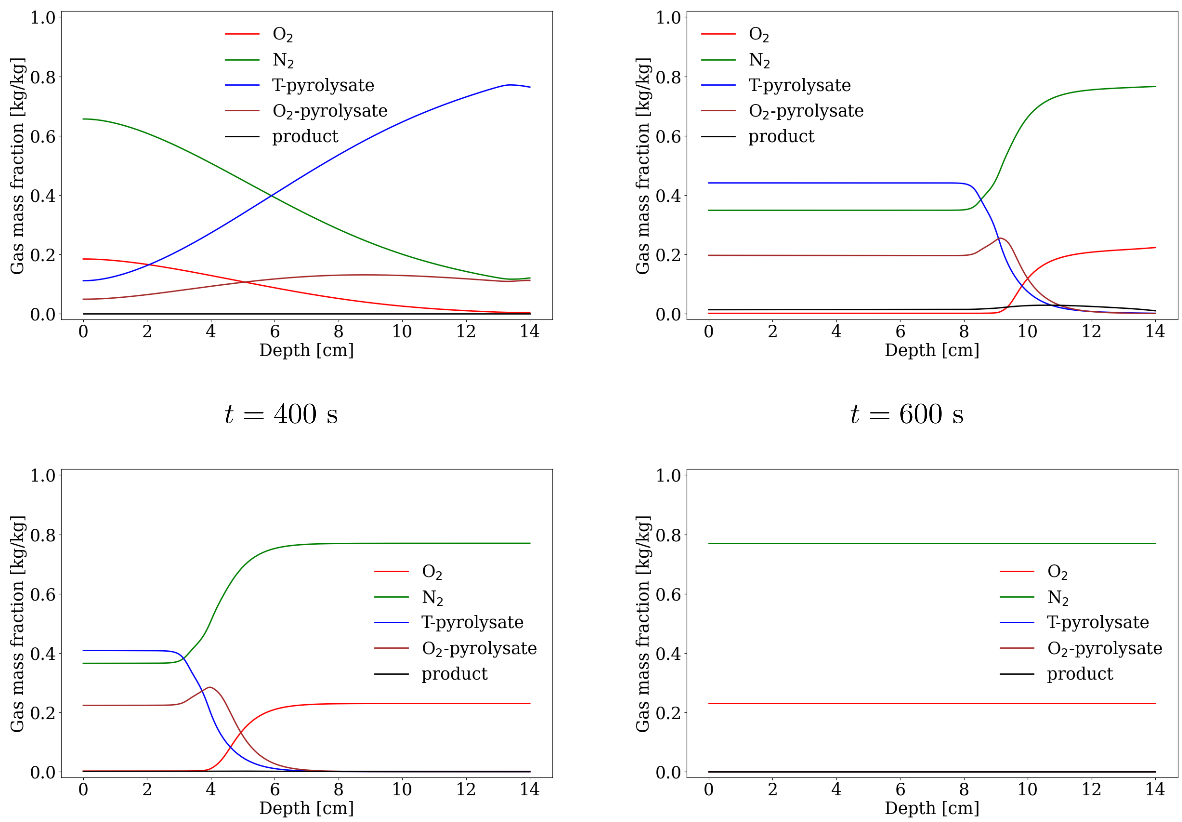
$t = 600$ s

$t = 800$ s

$t = 1000$ s

Figure 7.4: Solid-phase composition at $t = 400$, 600, 800, and 1000 s.

Gas-phase composition inside the sample. Figure 7.5 presents the gas-phase mass fractions inside the foam at the same four times. At $t = 600$ s and $t = 800$ s, oxygen is almost completely consumed within the smoldering front, as indicated by the oxygen mass fraction dropping to zero downstream of the reaction zone. This observation suggests that the thickness of the smoldering front is strongly controlled by the oxygen supply, with larger inlet fluxes expected to sustain wider reaction zones. Downstream of the front, the gas mixture is dominated by gaseous products of pyrolysis and oxidation. At $t = 1000$ s, once degradation has ceased, the remaining pyrolysis gases exit the sample, and the gas composition is that of the injected air.



$t = 400$ s

$t = 600$ s

$t = 800$ s

$t = 1000$ s

Figure 7.5: Gas-phase mass fractions inside the sample at $t = 400$, 600, 800, and 1000 s.

Smolder front propagation. Figure 7.6 focuses on the char mass fraction, which is a direct product of oxidation reactions (see reactions 7.4–7.5). Regions with non-zero char mass fraction clearly identify the active smoldering zone. The displacement of the char peak over time reveals a steady propagation of the smolder front along the sample. The propagation velocity is estimated by tracking the position of this peak, yielding a value of approximately 0.25 $\mathrm{mm\,s^{-1}}$. This value is consistent with experimentally reported measurements [9], supporting the validity of the numerical model.

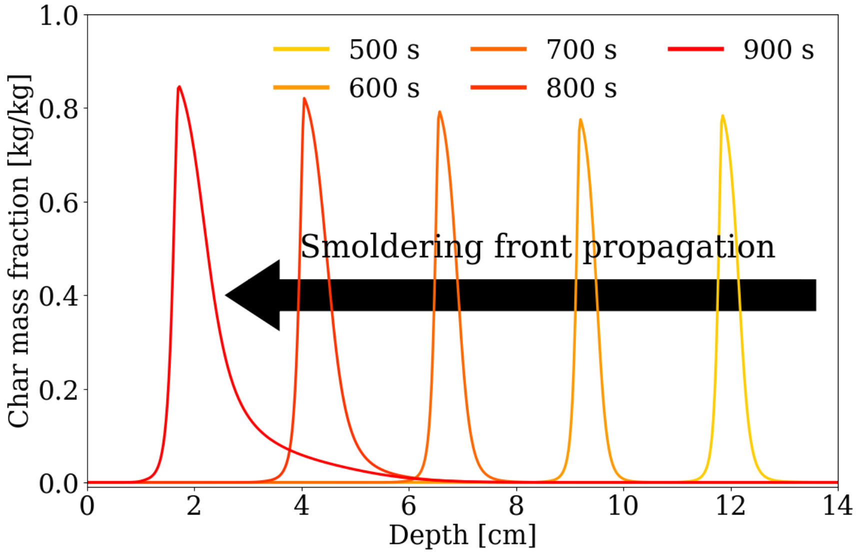


Figure 7.6: Smolder front propagation inferred from the spatial distribution of char mass fraction.

Gas mass fluxes. Figure 7.7 shows the net mass flux of gaseous species, defined as the difference between the outlet and inlet mass flow rates. The oxygen mass flux is negative, indicating oxygen consumption due to oxidation reactions occurring within the smoldering front. Although nitrogen is inert and does not participate in any reaction, a non-zero nitrogen mass flux is observed. This behavior is attributed to gas expansion caused by the temperature rise during degradation, which induces an outward flow of nitrogen driven by density variations.

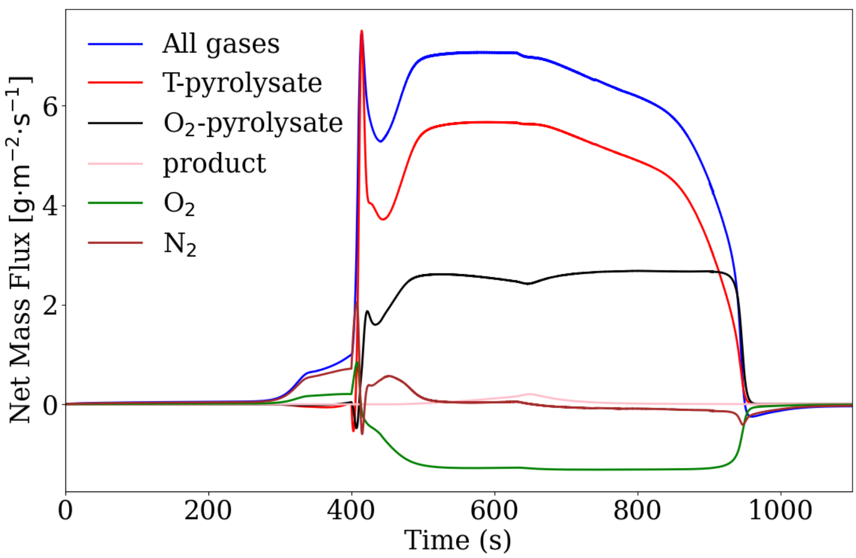


Figure 7.7: Net gaseous mass fluxes during the smoldering process.

Appendices

A Analytical solution of the 1D stationary pressure profile

A one–dimensional steady gas flow through a rigid porous medium is considered. The domain is a segment of length L, defined for $z \in [0, L]$. No volumetric gas sources are present, so the problem is strictly one–dimensional and steady.

The mass flux follows Darcy's law written for mass transport,

$$\dot{m}'' = \frac{\bar{K}}{\nu} \frac{dP}{dz}, \tag{9}$$

where $\bar{K}$ is the intrinsic permeability (m^2) and ν is the kinematic viscosity (m^2 s^{-1}). The viscosity is assumed inversely proportional to pressure,

$$\nu = \frac{\alpha}{P}, \tag{10}$$

which corresponds to the Chapman–Enskog dilute–gas formulation at constant temperature.

Mass conservation in steady state gives

$$\frac{d\dot{m}''}{dz} = 0, \tag{11}$$

implying a spatially uniform mass flux:

$$\dot{m}'' = \text{constant}. \tag{12}$$

Substituting (10) into Darcy's law yields

$$\dot{m}'' = \frac{\bar{K}P}{\alpha} \frac{dP}{dz}, \tag{13}$$

so the governing differential equation becomes

$$P \frac{dP}{dz} = \frac{\alpha}{\bar{K}} \dot{m}''. \tag{14}$$

A.1 Case 1: fixed pressure at $z = 0$ and $z = L$

The boundary conditions are

$$P(0) = P_0, \qquad P(L) = P_L. \tag{15}$$

Integrating (14) from $z = 0$ to $z = L$ gives

$$\frac{1}{2}\left(P_L^2 - P_0^2\right) = \frac{\alpha L}{\bar{K}} \dot{m}''. \tag{16}$$

The mass flux is therefore

$$\dot{m}'' = \frac{\bar{K}}{2\alpha L}\left(P_L^2 - P_0^2\right). \tag{17}$$

Using (14), the pressure profile becomes

$$P(z)^2 = P_0^2 + \left(P_L^2 - P_0^2\right) \frac{z}{L}. \tag{18}$$

A.2 Case 2: fixed pressure at $z = 0$ and imposed mass flux at $z = L$

The boundary conditions now read

$$P(0) = P_0, \qquad \dot{m}''(L) = \dot{m}''_L, \tag{19}$$

where $\dot{m}''_L$ is the prescribed flux at $z = L$. Since the flux is uniform [Eq. (12)], the entire domain satisfies

$$\dot{m}'' = \dot{m}''_L. \tag{20}$$

Inserting the imposed flux into (14) gives

$$P\,\frac{dP}{dz} = \frac{\alpha}{\bar{K}}\dot{m}''_L. \tag{21}$$

Integrating from 0 to z yields

$$P(z)^2 - P_0^2 = 2\,\frac{\alpha}{\bar{K}}\,\dot{m}''_L\,z. \tag{22}$$

The pressure profile for Case 2 therefore reads

$$P(z) = \sqrt{P_0^2 + 2\,\frac{\alpha}{\bar{K}}\,\dot{m}''_L\,z}. \tag{23}$$

If desired, the resulting pressure at $z = L$ follows directly as

$$P(L) = \sqrt{P_0^2 + 2\,\frac{\alpha L}{\bar{K}}\,\dot{m}''_L}. \tag{24}$$

This concludes the analytical pressure solutions for both a fully prescribed-pressure boundary configuration and a mixed pressure–flux boundary configuration.

B Analytical solution of mass conservation for Single-Step, First-Order non-charring reaction

A sample composed of a single condensed material A is considered. The material is assumed to be non-charring, and its thermal degradation produces only gaseous products, with no additional condensed species. Under these assumptions, the total mass of the sample m_{tot} is equal to the mass of material A, denoted m_A.

The general mass conservation equation for species A is written as:

$$\frac{dm_A}{dt} = \dot{\omega}'''_{\text{net},A}\,V \tag{25}$$

where V is the sample volume, defined as $V = m_{\text{tot}}/\bar{\rho}$, and $\dot{\omega}'''_{\text{net},A}$ is the net volumetric production rate of species A.

Since material A is only consumed by the degradation reaction, the net production rate reduces to:

$$\dot{\omega}'''_{\text{net},A} = -\dot{\omega}'''_{\text{dest},A} = -r_k(T) \tag{26}$$

where $r(T)$ is the rate of the single reaction responsible for the destruction of material A. The reaction is assumed to be non-oxidative and of first order.

The reaction rate is expressed as:

$$r(T) = (1 - \alpha_A)\,\frac{\bar{\rho} Y_A}{1 - \alpha_A}\,K(T) = \bar{\rho} Y_A K(T) \tag{27}$$

where α_A is the conversion fraction of material A, Y_A is its mass fraction, and $K(T)$ is the temperature-dependent reaction rate constant.

The reaction rate constant follows an Arrhenius law:

$$K(T) = Z \exp\left(-\frac{E}{RT}\right) \tag{28}$$

where Z is the pre-exponential factor, E is the activation energy, R is the universal gas constant, and T is the absolute temperature.

The configuration is analogous to a thermogravimetric analysis (TGA), and the temperature evolution is prescribed as a linear ramp:

$$T(t) = T_0 + \beta t \tag{29}$$

where T_0 is the initial temperature and β is the heating rate.

Since material A is the only condensed species, its mass fraction remains equal to unity ($Y_A = 1$). Equation 25 therefore simplifies to:

$$\frac{dm_A}{dt} = -K(T)\,m_A = -Z \exp\left(-\frac{E}{RT(t)}\right) m_A \tag{30}$$

This expression corresponds to a first-order kinetic reaction with a known analytical solution. The mass of material A as a function of temperature is given by:

$$m_A(t) = m_{A,0} \exp\left(-\int_{t_0}^{t} Z \exp\left(-\frac{E}{R\,(T_0 + \beta t')}\right)\,dt'\right) \tag{31}$$

where $m_{A,0}$ is the initial mass of the sample.

The mass loss rate (MLR) is defined as:

$$\mathrm{MLR}(t) = -\frac{dm_A}{dt} \tag{32}$$

Substituting the Arrhenius expression and the integrated mass yields:

$$\mathrm{MLR}(t) = m_{A,0} Z \exp\left(-\frac{E}{R\,(T_0 + \beta t)}\right)\,\exp\left(-\int_{t_0}^{t} Z \exp\left(-\frac{E}{R\,(T_0 + \beta t')}\right)\,dt'\right) \tag{33}$$

This expression may also be written in a compact form as:

$$\mathrm{MLR}(T) = m_{A,0}\,K(T)\,\exp\left(-\int_{T_0}^{T} \frac{K(\tau)}{\beta}\,d\tau\right) \tag{34}$$

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